

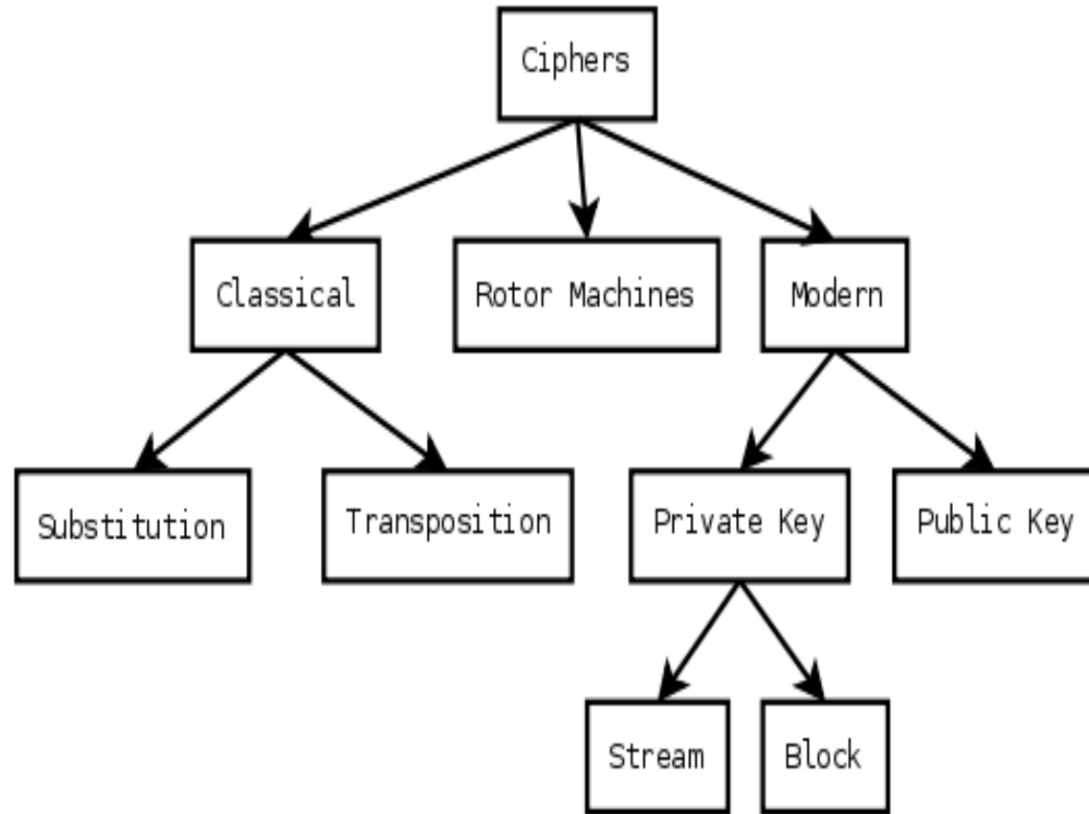
Unit 2: Cryptography

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Basic Terminology

- **Plaintext** - the original message
- **Ciphertext** - the coded message
- **Cipher** - algorithm for transforming plaintext to ciphertext
- **Key** - info used in cipher known only to sender/receiver
- **Encipher (encrypt)** - converting plaintext to ciphertext
- **Decipher (decrypt)** - recovering plaintext from ciphertext
- **Cryptography** - study of encryption principles/methods
- **Cryptanalysis (codebreaking)** - the study of principles/methods of deciphering ciphertext *without* knowing key
- **Cryptology** - the field of both cryptography and cryptanalysis

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- **Can characterize by:**
 - **Type of encryption operations used**
 - Substitution / Transposition / Product
 - **Number of keys used**
 - Single-key or Private Key (Symmetric Encryption)
 - Two-key or Public Key (Asymmetric encryption)
 - **Way in which plain text is processed**
 - Block / stream
 - Block cipher processes the input one block of elements at a time.
 - Stream cipher processes the input elements continuously, producing output one element at a time, as it goes along.

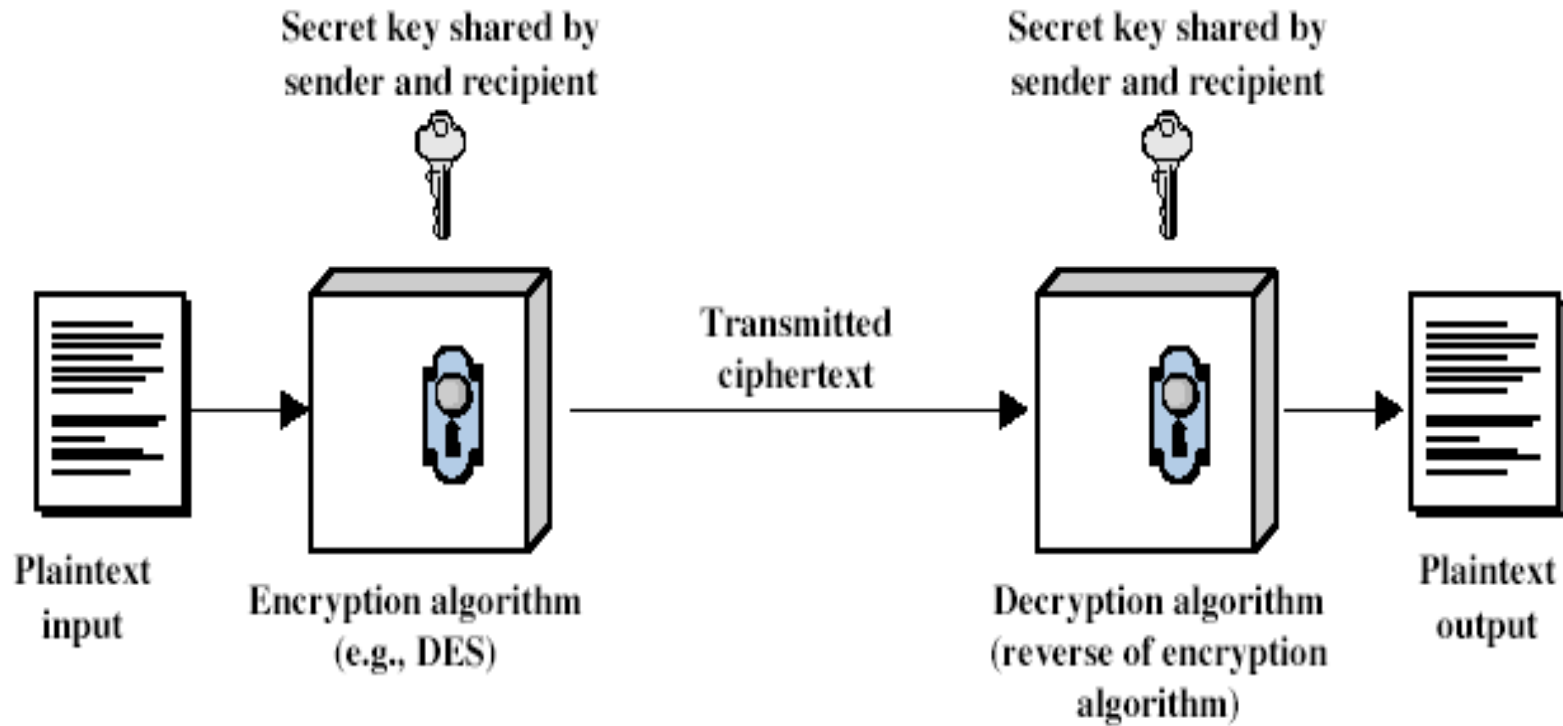
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Symmetric Encryption

- Or conventional / private-key / single-key
- Sender and recipient share a common key
- All classical encryption algorithms are private-key
- Was only type prior to invention of public-key in 1970's

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Symmetric Cipher Model



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Requirements

- **Two requirements for secure use of symmetric encryption:**
 - A strong encryption algorithm
 - A secret key known only to sender / receiver

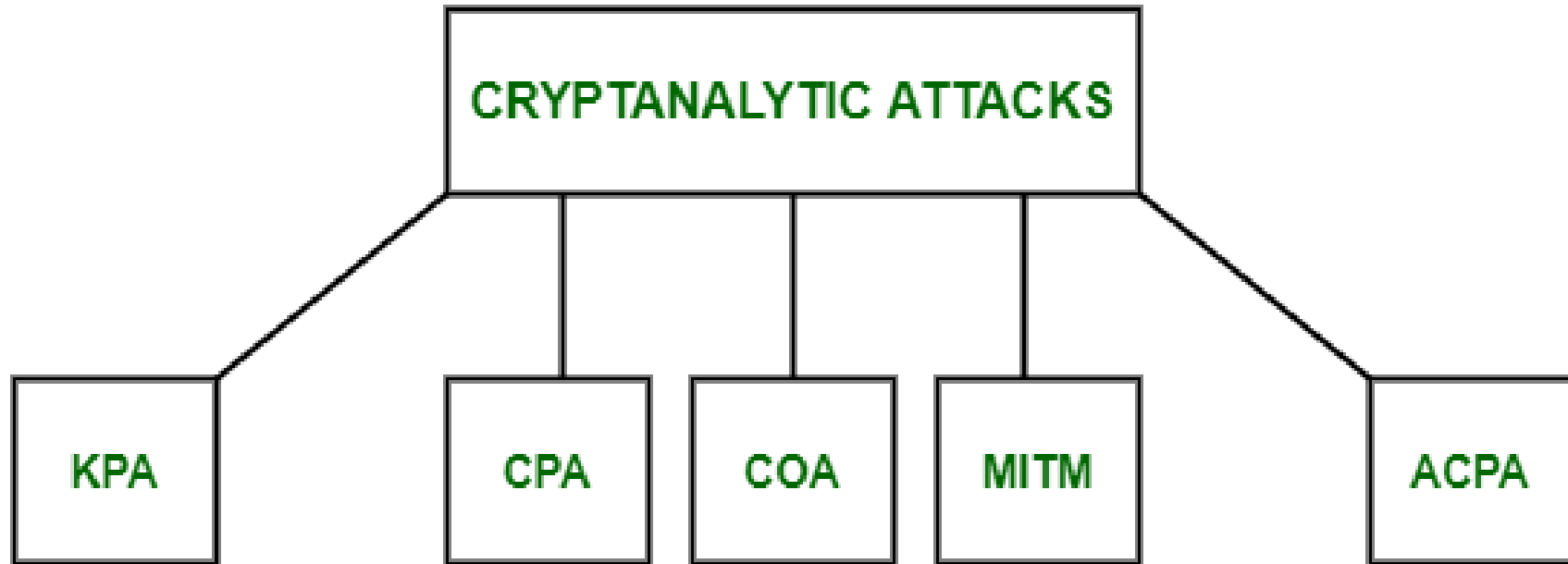
$$CT = E_k(PT)$$

$$PT = D_k(CT)$$

- **Assume encryption algorithm is known**
- **Implies a secure channel to distribute key**

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Types of Cryptanalytic Attacks



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Types of Cryptanalytic Attacks

- **Known-Plaintext Analysis (KPA):** In this type of attack, some plaintext-ciphertext pairs are already known. Attacker maps them in order to find the encryption key. This attack is easier to use as a lot of information is already available.
- **Chosen-Plaintext Analysis (CPA):** In this type of attack, the attacker chooses random plaintexts and obtains the corresponding ciphertexts and tries to find the encryption key. Its very simple to implement like KPA but the success rate is quite low.

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Types of Cryptanalytic Attacks

- **Ciphertext-Only Analysis (COA):** In this type of attack, only some cipher-text is known and the attacker tries to find the corresponding encryption key and plaintext. Its the hardest to implement but is the most probable attack as only ciphertext is required.
- **Man-In-The-Middle (MITM) attack:** In this type of attack, attacker intercepts the message/key between two communicating parties through a secured channel.

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Types of Cryptanalytic Attacks

- **Adaptive Chosen-Plaintext Analysis (ACPA):** This attack is similar CPA. Here, the attacker requests the cipher texts of additional plaintexts after they have ciphertexts for some texts.

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Brute Force Search

- Always possible to simply try every key
- Most basic attack, proportional to key size
- Assume either know / recognise plaintext

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Brute Force Search

Key Size (bits)	Number of Alternative Keys	Time required at 1 decryption/ μ s	Time required at 10^6 decryptions/ μ s
32	$2^{32} = 4.3 \times 10^9$	$2^{31} \mu$ s = 35.8 minutes	2.15 milliseconds
56	$2^{56} = 7.2 \times 10^{16}$	$2^{55} \mu$ s = 1142 years	10.01 hours
128	$2^{128} = 3.4 \times 10^{38}$	$2^{127} \mu$ s = 5.4×10^{24} years	5.4×10^{18} years
168	$2^{168} = 3.7 \times 10^{50}$	$2^{167} \mu$ s = 5.9×10^{36} years	5.9×10^{30} years
26 characters (permutation)	$26! = 4 \times 10^{26}$	$2 \times 10^{26} \mu$ s = 6.4×10^{12} years	6.4×10^6 years

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Definitions

- **Unconditional security**
 - No matter how much computer power is available, the cipher cannot be broken since the ciphertext provides insufficient information to uniquely determine the corresponding plaintext
- **Computational security**
 - Given limited computing resources (eg time needed for calculations is greater than age of universe), the cipher cannot be broken

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Classical Substitution Ciphers

- Where letters of plaintext are replaced by other letters or by numbers or symbols
- Or if plain text is viewed as a sequence of bits, then substitution involves replacing plaintext bit patterns with ciphertext bit patterns

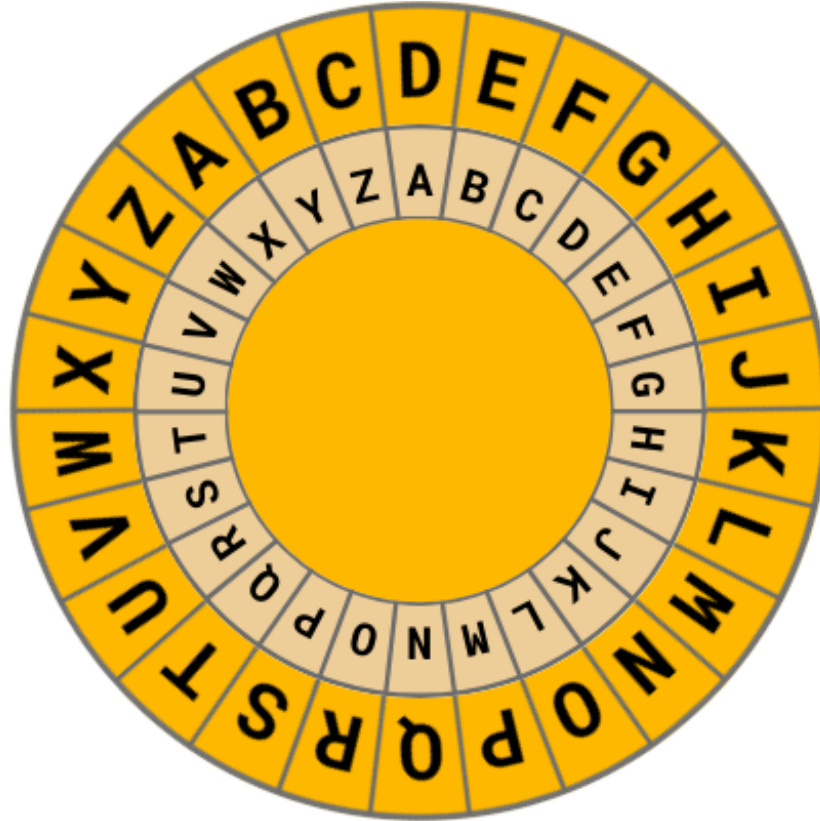
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Caesar Cipher

- Earliest known substitution cipher
- By Julius Caesar
- First attested use in military affairs
- Replaces each letter by 3rd letter on
- Example:
 - meet me after the toga party
 - PHHW PH DIWHU WKH WRJD SDUWB

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Caesar Cipher



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Caesar Cipher

- can define transformation as:
 - a b c d e f g h i j k l m n o p q r s t u v w x y z
 - D E F G H I J K L M N O P Q R S T U V W X Y Z A B C
- mathematically give each letter a number

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
u	v	w	x	y	z														
20	21	22	23	24	25														
- then have Caesar cipher as:
 - $C = E(p) = (p + k) \bmod (26)$
 - $p = D(C) = (C - k) \bmod (26)$

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Cryptanalysis of Caesar Cipher

- Only have 26 possible ciphers
 - A maps to a, b, ... Z
- Could simply try each in turn
- A **Brute Force Search**
- Given ciphertext, just try all shifts of letters
- Do need to recognize when have plain text
- Eg. break ciphertext "gcua vq dtgcm"

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Monoalphabetic Cipher

- Rather than just shifting the alphabet
- Could shuffle (jumble) the letters arbitrarily
- Each plaintext letter maps to a different random ciphertext letter
- Hence key is 26 letters long
 - Plain: abcdefghijklmnopqrstuvwxyz
 - Cipher: DKVQFIBJWPESCXHTMYAUOLRGZN
- Plaintext: ifwewishtoreplaceletters
- Ciphertext: WIRFRWAJUHYFTSDVFSFUUFYA

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Monoalphabetic Cipher

- Now have a total of $26! = 4 \times 10^{26}$ keys
- With so many keys, might think is secure
- But would be **!!!WRONG!!!**
- Problem is language characteristics

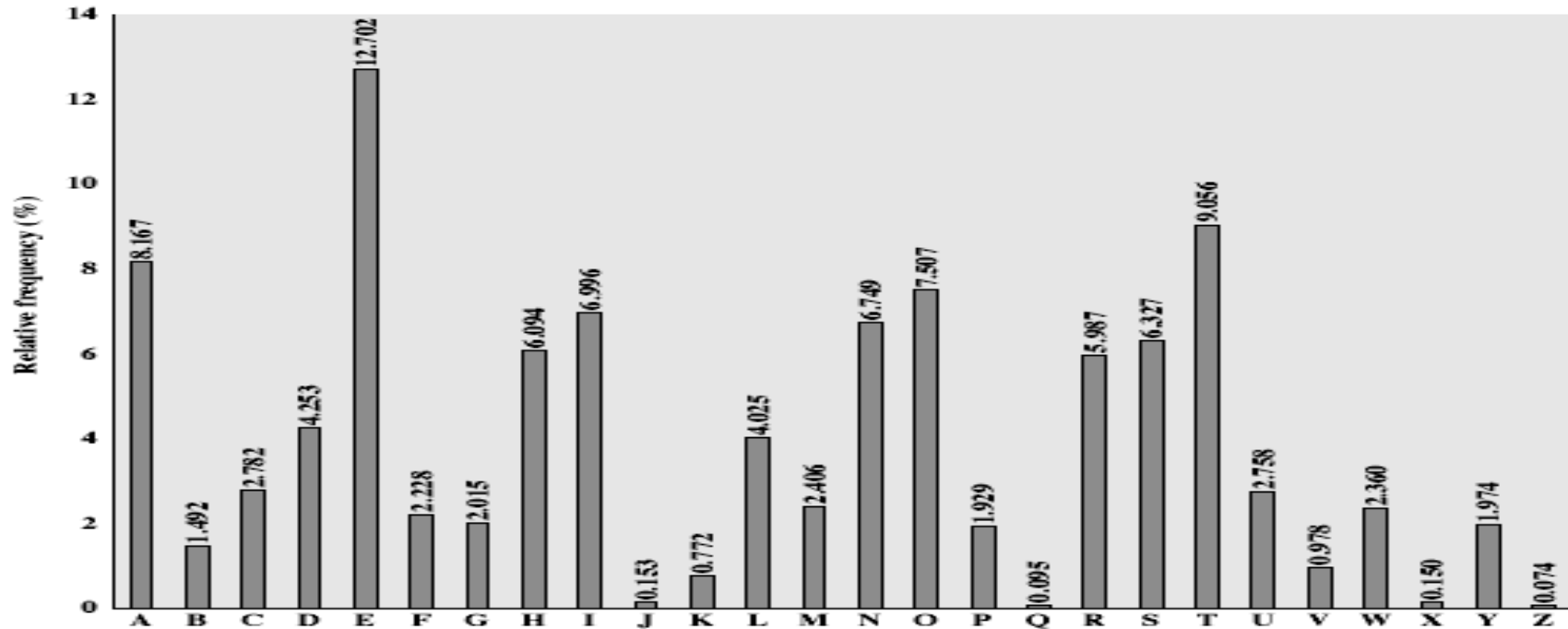
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Language Redundancy and Cryptanalysis

- Human languages are **redundant**
- Eg "th lrd s m shphrd shll nt wnt"
- Letters are not equally commonly used
- In english **e** is by far the most common letter
- Then t,r,n,i,o,a,s
- Other letters are fairly rare
- Cf. Z,j,k,q,x
- Have tables of single, double & triple letter frequencies

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English Letter Frequencies



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Use in Cryptanalysis

- Key concept - monoalphabetic substitution ciphers do not change relative letter frequencies
- Discovered by arabian scientists in 9th century
- Calculate letter frequencies for ciphertext
- Compare counts/plots against known values
- If caesar cipher look for common peaks/troughs
 - Peaks at: a-e-i triple, no pair, rst triple
 - Troughs at: jk, x-z
- For monoalphabetic must identify each letter
 - Tables of common double/triple letters help

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Example Cryptanalysis

- given ciphertext:
 - UZQSOVUOHXMOPVGPOZPEVSGZWSZOPFPESXUDBMETSXAIZ
 - VUEPHZHMDZSHZOWSFPAPPDTSVPQUZWMYXUZHUSX
 - EPYEPOPDZSZUFPOMBZWPFUPZHMDJUDTMOHMQ
- count relative letter frequencies (see text)
- guess P & Z are e and t
- guess ZW is th and hence ZWP is the
- proceeding with trial and error finally get:
 - it was disclosed yesterday that several informal but
 - direct contacts have been made with political
 - representatives of the viet cong in moscow

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Playfair Cipher

- Not even the large number of keys in a monoalphabetic cipher provides security
- One approach to improving security was to encrypt multiple letters
- The **playfair cipher** is an example
- Invented by charles wheatstone in 1854, but named after his friend baron playfair

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Playfair Key Matrix

- A 5x5 matrix of letters based on a keyword
- Fill in letters of keyword (sans duplicates)
- Fill rest of matrix with other letters
- Eg. using the keyword MONARCHY

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

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Encrypting and Decrypting

- Plaintext encrypted two letters at a time:
 - If a pair is a repeated letter, insert a filler like 'x', eg. "balloon" encrypts as "ba lx lo on"
 - If both letters fall in the same row, replace each with letter to right (wrapping back to start from end), eg. "ar" encrypts as "rm"
 - If both letters fall in the same column, replace each with the letter below it (again wrapping to top from bottom), eg. "mu" encrypts to "cm"
 - Otherwise each letter is replaced by the one in its row in the column of the other letter of the pair, eg. "hs" encrypts to "bp", and "ea" to "im" or "jm" (as desired)

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Encrypting and Decrypting

- Split the plaintext first into digraphs (pairs of two letters). If the plaintext has an odd number of letters, append the letter Z at the end. Even the text in the basic form of The MANGO plaintext, for instance, consists of five letters. Consequently, it is not possible to make a digraph. As a result, we will change the plaintext to MANGOZ by adding the letter Z at the end.
- Divide the plaintext into digraphs after that (pair of two letters). For any letter that appears twice, place an X there (side by side). Think about the scenario where the digraph is CO MX MU NI CA TE and the plaintext is COMMUNICATE. For plaintext GREET, the digraph will be GR EX ET, but for plaintext JAZZ, the digraph will be JA ZX ZX.

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Example of Playfair Cipher

- Let's say we want to encrypt the message: "HELLO WORLD" using the key phrase "KEYWORD."

Generating the Key Table:

- Step 1:** Create a key table by using the key phrase "KEYWORD" and filling it with unique letters (excluding duplicates) in a specific order. The remaining letters of the alphabet are then added in order, omitting the letters already present in the key phrase. The letter "I" is usually combined with "J" in the same grid cell.

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Example of Playfair Cipher

- The key matrix for our example would look like this:

K	E	Y	W	O
R	D	A	B	C
F	G	H	I/J	L
M	N	P	Q	S
T	U	V	X	Z

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Example of Playfair Cipher

- **Step 2 - Preparing the Message:**

The next step is to prepare the message for encryption. We remove any non-alphabetic characters and convert all letters to uppercase. In our example, the message "HELLO WORLD" becomes "HELLOWORLD".

- **Step 3 - Breaking the Message into Pairs:** To encrypt the message using the Playfair cipher, we break the message into pairs of letters. If we have an odd number of letters, we usually add a placeholder letter (e.g., "X") at the end to create a pair. If in between duplicate letters come in same pair then we will break it into single letter and attach X at the first letter then will start making the other pairs. In our example, "HELLOWORLD" becomes "HE | LX | LO | WO | RL | DX".

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Example of Playfair Cipher

- **Step 4:** Now take one pair at a time and look for it in the key matrix.
 - **HE:** If we look for “HE” in the key matrix it makes 3 X 2 matrix. So, we will look in row of “H” and column of “E” there intersection will be encryption of “H”. “G” becomes the answer for “H”. Now look in row of “E” and column of “H”. There intersection will become the encryption of “E”. “Y” becomes the answer for “E”. So, “HE” becomes “GY”.
 - **LX:** If we look for “LX” in the key matrix it makes 3 X 2 matrix. So, we will look in row of “L” and column of “X” there intersection will be encryption of “L”. Either “I” or “J” can be taken. We will take “J” which becomes the answer for “L”. Now look in row of “X” and column of “L”. There intersection will become the encryption of “X”. “Z” becomes the answer for “X”. So, “LX” becomes “JZ”.

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Example of Playfair Cipher

- **LO:** For “LO” it doesn’t form any matrix in key matrix because they are appearing in same column. For this type of situation we will move one character to the down for encryption of “L”. So, “S” will become encrypted character for “L”. for “O” also we will move one character down. “C” will become answer for “O”. So, “LO” becomes “SC”.
- **WO:** For “WO” it doesn’t form any matrix in key matrix because they are appearing in same row. For this type of situation we will move one character to the right for encryption of “W”. So, “O” will become encrypted character for “W”. for “O” also we will move one character right. (If it is end of the row/column start from the 1st element of row/column) “K” will become answer for “O”. So “WO” becomes “OK”.

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Example of Playfair Cipher

- **RL:** If we look for “RL” in the key matrix it makes 2 X 5 matrix. So, we will look in row of “R” and column of “L” there intersection will be encryption of “R”. “C” becomes the answer for “R”. Now look in row of “L” and column of “R”. There intersection will become the encryption of “L”. “F” becomes the answer for “L”. So, “RL” becomes “CF”.
- **DX:** If we look for “DX” in the key matrix it makes 4 X 3 matrix. So, we will look in row of “D” and column of “X” there intersection will be encryption of “D”. “B” becomes the answer for “D”. Now look in row of “X” and column of “D”. There intersection will become the encryption of “X”. “U” becomes the answer for “X”. So, “DX” becomes “BU”.

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Example of Playfair Cipher

- Step 5: Finally the encrypted answer for “HELXLOWORLDX” is “GYJZSCOKCFBU”.
- Now the same procedure can be reversed for finding the original message from given cipher text and then remove extra “X” from it.

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Security of Playfair Cipher

- Security much improved over monoalphabetic
- Since have $26 \times 26 = 676$ diagrams
- Would need a 676 entry frequency table to analyse (verses 26 for a monoalphabetic)
- And correspondingly more ciphertext
- Was **widely used** for many years (eg. Us & british military in ww1)
- It **can be broken**, given a few hundred letters
- Since still has much of plaintext structure

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Playfair Cipher Restrictions

- Only 25 alphabets are supported.
- It is incompatible with characters that number.
- Only capital or lowercase letters are accepted.
- Special characters like spaces, newlines, punctuation, etc. are not allowed.

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Hill Ciphers

- Developed by mathematician Lester Hill in 1929.
- M successive plaintext is substituted by m cipher text letters.
- Each character is assigned a numeric value($a=0, b=1, \dots, z=25$)

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Hill Ciphers

- For $m=3$

$$\begin{pmatrix} C1 \\ C2 \\ C3 \end{pmatrix} = \begin{pmatrix} K11 & K12 & K13 \\ K21 & K22 & K23 \\ K31 & K32 & K33 \end{pmatrix} \begin{pmatrix} P1 \\ P2 \\ P3 \end{pmatrix} \text{ mod } 26$$

- $C1 = ((K11*P1) + (K12*P2) + (K13*P3)) \text{ mod } 26$
- $C2 = ((K21*P1) + (K22*P2) + (K23*P3)) \text{ mod } 26$
- $C3 = ((K31*P1) + (K32*P2) + (K33*P3)) \text{ mod } 26$
 $C = KP \text{ mod } 26$

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Hill Ciphers

- Eg: pay
- Pay = 15 0 24
K = 17 17 5
21 18 21
2 2 19
- $C1 = (17 * 15) + (17 * 0) + (5 * 24) = 255 + 0 + 120 = 375$
- $C2 = (21 * 15) + (18 * 0) + (21 * 24) = 315 + 0 + 504 = 819$
- $C3 = (2 * 15) + (2 * 0) + (19 * 24) = 30 + 0 + 456 = 486$
- $C = (375 \ 819 \ 486) \bmod 26$
- $C = (11 \ 13 \ 18) = \text{LNS}$

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Polyalphabetic Ciphers

- Another approach to improving security is to use multiple cipher alphabets
- Called **polyalphabetic substitution ciphers**
- Makes cryptanalysis harder with more alphabets to guess and flatter frequency distribution
- Use a key to select which alphabet is used for each letter of the message
- Use each alphabet in turn
- Repeat from start after end of key is reached

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Vigenère Cipher

- Simplest polyalphabetic substitution cipher
- Write the plaintext out
- Write the keyword repeated above it
- Use each key letter as a caesar cipher key
- Encrypt the corresponding plaintext letter

Plaintext

Key

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	D
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	C
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	B
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

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Example of Vigenère Cipher

- Eg using keyword *deceptive*
 - key: deceptivedeceptivedeceptive
 - plaintext: wearediscoveredsaveyourself
 - ciphertext: ZICVTWQNGRZGVTWAVZHCQYGLMGJ

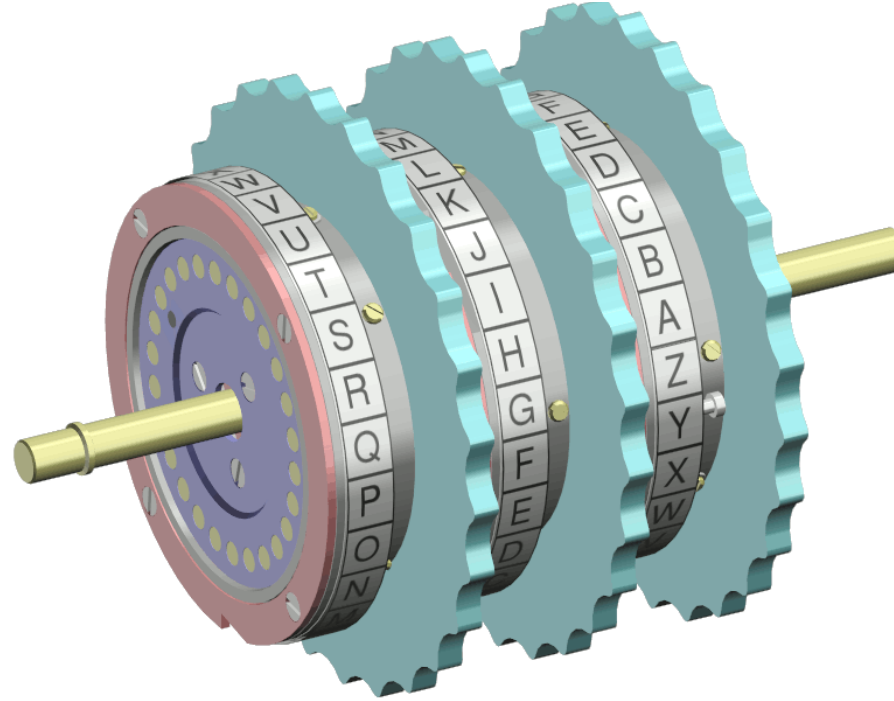
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Rotor Machines

- A rotor machine is an electro-mechanical encryption device used for secure communication. It consists of rotating disks called rotors that create a substitution cipher. When a letter is typed, an electrical current passes through the rotors, causing them to rotate and encrypt the letter. Decryption is performed by inputting the encrypted message and reversing the process. Rotor machines, like the Enigma machine, were used during World War II and provided a high level of security. However, they have been surpassed by modern encryption methods.

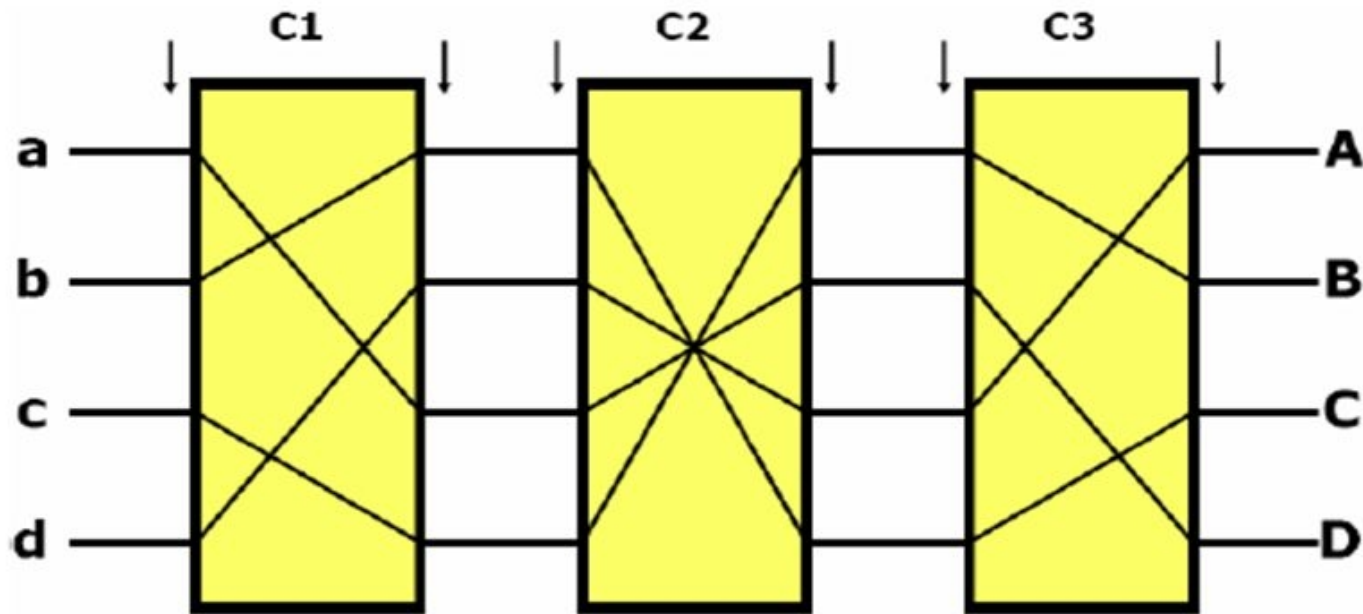
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Rotor Machines



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Rotor Machines



This setting maps: $a \rightarrow D$, $b \rightarrow C$, $c \rightarrow B$, $d \rightarrow A$

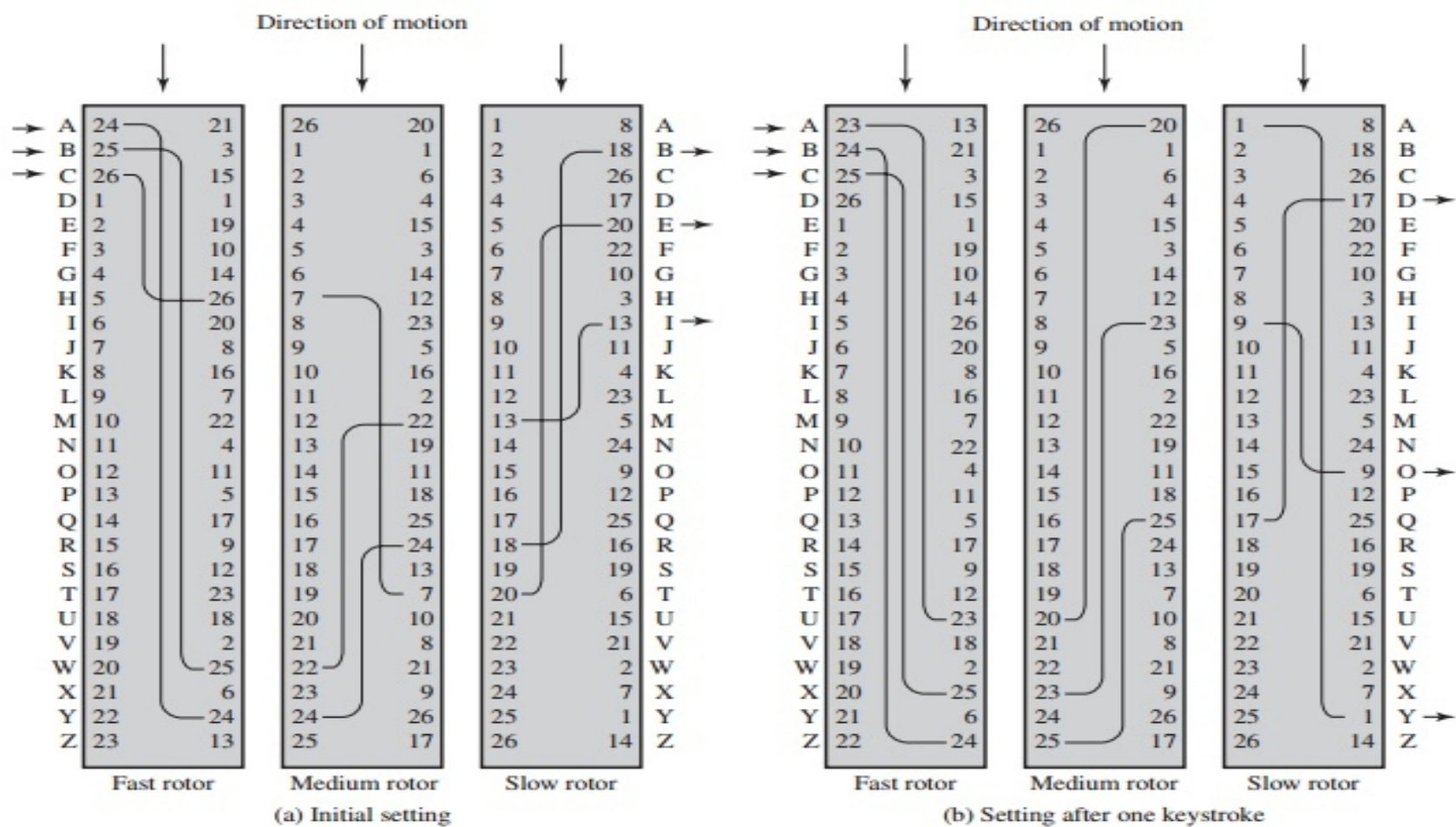


Figure 2.8 Three-Rotor Machine with Wiring Represented by Numbered Contacts

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Rotor Machine Example

- Let's take a simplified example of a rotor machine encryption process using three rotors.
- Rotor Setup:
 - We have three rotors, labeled R1, R2, and R3, each with 26 electrical contacts on each side representing the letters of the alphabet.
 - The rotors are placed on an axle and can be rotated manually or automatically.
 - Each rotor has a specific wiring configuration, which determines the substitution cipher.

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Rotor Machine Example

- Encryption Process:
 - Suppose we want to encrypt the letter "A" using the rotor machine.
 - The letter "A" is pressed on the keyboard, and an electrical current passes through the rotors.
 - The current enters rotor R1, goes through the contacts, and exits at a different letter, let's say "H".
 - The current then enters rotor R2, goes through the contacts, and exits at another letter, such as "X".
 - Finally, the current enters rotor R3, goes through the contacts, and exits at the encrypted letter, for example, "L".
 - The encrypted letter "L" is then displayed or transmitted as the ciphertext.

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Rotor Machine Example

- Decryption Process:
 - To decrypt the ciphertext "L" back to plaintext "A", the encrypted message is entered into the rotor machine.
 - The electrical current passes through rotor R3 in reverse, following the contacts and exiting at a letter, say "Q".
 - The current then enters rotor R2, goes through the contacts, and exits at another letter, such as "B".
 - Finally, the current enters rotor R1, goes through the contacts, and exits at the decrypted letter, "A".
 - The decrypted letter "A" is then displayed or used as needed.
- Please note that this example is simplified for illustrative purposes. In reality, rotor machines were more complex, employing additional components such as a plugboard for extra encryption and intricate rotor settings to enhance security.

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Steganography

- Steganography comes from the greek word which means covered writing.
- Steganography is the practice of hiding secret information within a seemingly innocuous carrier, such as an image, audio file, video, or even text, without raising suspicion. It is often used in conjunction with cryptography to provide an additional layer of security.
- While cryptography focuses on encrypting and decrypting data to protect its confidentiality, steganography focuses on concealing the very existence of the hidden information. By embedding the secret data within the carrier medium, steganography aims to make the presence of the secret information undetectable to unauthorized individuals.

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Steganography

- Here's a high-level overview of how steganography can be used with cryptography:
- **Encryption of Secret Data:**
 - First, the original secret data that needs to be protected is encrypted using cryptographic algorithms.
 - The encryption process ensures that the secret data is scrambled and transformed into an unreadable format, known as ciphertext.
- **Selection of Carrier Medium:**
 - A carrier medium, such as an image or audio file, is chosen as the cover for hiding the encrypted secret data.
 - The carrier medium should ideally have a large amount of redundant or unnoticeable data that can be modified without significantly altering its appearance or quality.

Unit 2: Cryptography

Steganography

- **Embedding the Secret Data:**
 - The encrypted secret data is then embedded or concealed within the carrier medium.
 - This can be achieved by subtly modifying the carrier's data in specific ways that only authorized recipients with the proper knowledge or tools can detect and extract the hidden information.
 - The modifications are designed to be imperceptible to the human eye or ear, ensuring that the carrier medium appears unchanged.

Unit 2: Cryptography

Steganography

- **Transmission and Extraction:**

- The modified carrier medium, now containing the embedded secret data, can be transmitted openly without raising suspicion.
- Upon reaching the intended recipient, the extraction process takes place.
- The recipient, possessing the knowledge or tools required, extracts the hidden information from the carrier medium.
- The extracted information is then decrypted using cryptographic techniques to retrieve the original secret data.

By combining steganography and cryptography, sensitive information can be hidden in plain sight, making it more challenging for unauthorized individuals to detect or intercept the secret data. However, it's worth noting that steganography **does not provide encryption** itself but rather **focuses on the covert transmission of encrypted data**.

Unit 2: Cryptography

Steganography

- **Types of Steganography**
 - **Text Staganography** : Text steganography can be achieved by altering the text formatting, or by altering certain characteristics of textual elements (e.g., characters).
 - **Image Staganography** : Images are used as the message carriers. Images are the most popular cover objects used for steganography.
 - **Audio Staganography** : In audio steganography, secret message is embedded into digitized audio signal which result slight altering of binary sequence of the corresponding audio file.

Unit 2: Cryptography

Steganography Examples

- 1 A hidden message that can be revealed by playing a video at a faster frame rate
- 2 Content that can be unveiled by playing an audio track backward
- 3 Embedding a message into an RGB image by hiding it in the red, green, or blue channel
- 4 Adding noise or sound to encrypt a message within a photo
- 5 Hiding content within the file header or metadata
 - A simpler example is a message in plain text. For example, in the following sentence, the first letter of each word reveals the word hidden:
 - Harriette is definitely done entertaining nonsense.

Unit 2: Cryptography

Steganography Examples

- **Character marking**
 - Selected letters of printed or typewritten text are over written in pencil. The marks are ordinarily not visible unless the paper is held at an angle to bright light.
- **Invisible ink**
 - A number of substances can be used for writing but leave no visible trace until heat or some chemical is applied to the paper.
- **Pin punctures**
 - Small pin punctures on selected letters are ordinarily not visible unless the paper is held up in front of a light.
- **Type writer correction ribbon**
 - Used between lines typed with a black ribbon, the results of typing with the correction tape are visible only under a strong light.

Unit 2: Cryptography

Types of Security

Theoretical Security (Provable Security):

- Relies on mathematical proofs to demonstrate the security of a cryptographic algorithm under specific assumptions.
- Provides strong guarantees about the algorithm's security if the underlying assumptions hold true.
- Does not consider practical limitations or computational resources.
- Offers a high level of confidence in security but may not account for real-world constraints.

Unit 2: Cryptography

Computational Security (Practical Security):

- Focuses on real-world security by considering the limitations of computational resources available to attackers.
 - Evaluates the security of a cryptographic algorithm based on the assumption that certain computational problems are difficult to solve efficiently.
 - Aims to ensure that breaking the algorithm requires more resources than an attacker can feasibly obtain.
 - Provides a practical approach but lacks the same level of mathematical certainty as theoretical security.
-
- In practice, cryptographic algorithms should strive to achieve both theoretical security, with strong mathematical guarantees, and computational security, by being resistant to attacks given real-world computational limitations. A good cryptographic system finds a balance between these two aspects to ensure security in practical scenarios.

Unit 2: Cryptography

Product Cryptosystem

- A product cryptosystem is a cryptographic construction that combines multiple cryptographic primitives or schemes to achieve specific security properties or functionalities. It's essentially a way of composing different cryptographic tools together to enhance security or address specific requirements.

Unit 2: Cryptography

Product Cryptosystem

- The motivations for using product cryptosystems may include:
- **Enhanced Security:** Combining different cryptographic primitives can create a more robust and secure system. For example, using a combination of encryption schemes and digital signatures can provide confidentiality and integrity simultaneously.
- **Flexibility:** Product cryptosystems can be designed to be more adaptable to different scenarios or environments, allowing for greater flexibility in security configurations.

Unit 2: Cryptography

Product Cryptosystem

- **Customization:** By combining cryptographic primitives, it is possible to tailor the resulting system to specific use cases or requirements, providing a customized solution.
- **Reducing Single-Point Vulnerabilities:** Using a single cryptographic scheme can lead to a single point of failure. By employing product cryptosystems, vulnerabilities in one component may not compromise the entire system.

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Unit 2: Cryptography

Product Cryptosystem

- **Compliance:** Certain cryptographic standards or regulations may require the use of specific combinations of cryptographic primitives. Employing a product cryptosystem can help organizations meet these compliance requirements.
- **Future-Proofing:** Cryptographic technologies evolve rapidly, and what is secure today may become vulnerable in the future. A product cryptosystem can integrate different schemes to provide better resistance against potential future attacks.

Unit 2: Cryptography

Product Cryptosystem

- Product cryptosystems are complicated and require extensive research and evaluation of the security features of the components and their interactions. Correct implementation and configuration of the pieces also affect system security. Thus, cryptographers and security experts should build and use product cryptosystems.

Unit 2: Cryptography

Symmetric key cryptography

- Symmetric key cryptography, also known as secret key cryptography or conventional cryptography, is a type of cryptographic system where the same secret key is used for both the encryption and decryption of data. It is one of the two primary categories of encryption, the other being asymmetric key cryptography (public-key cryptography).

Unit 2: Cryptography

Symmetric key cryptography

- In symmetric key cryptography, the key used to encrypt the data is kept secret and known only to the parties involved in the communication. The strength and security of the system depend on the secrecy of the key. If an unauthorized entity gains access to the secret key, they can easily decrypt the ciphertext and compromise the confidentiality of the data.

Unit 2: Cryptography

Symmetric key cryptography

- **Key Generation:**

Before any data can be encrypted or decrypted, a secret key needs to be generated. The key is usually a random string of bits with a specific length, depending on the chosen symmetric encryption algorithm.

- **Encryption:**

The sender uses the secret key and a symmetric encryption algorithm to process the plaintext data (original data) into ciphertext (encrypted data).

The ciphertext appears random and indecipherable without the secret key.

Unit 2: Cryptography

Symmetric key cryptography

- **Decryption:**

The receiver, who possesses the same secret key, uses it along with the symmetric encryption algorithm to decrypt the ciphertext and recover the original plaintext.

Unit 2: Cryptography

Characteristics Symmetric key cryptography

- **Speed and Efficiency:** Symmetric key encryption is generally faster and more efficient than asymmetric encryption since the same key is used for both encryption and decryption.
- **Key Management:** Key distribution and management are crucial in symmetric key cryptography. Both parties must securely share the secret key before they can communicate securely. Key exchange protocols like Diffie-Hellman can be used to establish a shared secret key without directly transmitting it.

Unit 2: Cryptography

Characteristics Symmetric key cryptography

- **Limited Scalability:** Symmetric encryption faces challenges in large-scale communication scenarios where many parties need to securely communicate with each other. The distribution and management of secret keys become increasingly complex as the number of parties grows.
- **Security Concerns:** The security of symmetric encryption relies on the key's secrecy. If an attacker can obtain or guess the secret key, they can decrypt the data. This concern becomes more pronounced as the number of parties involved in communication increases.

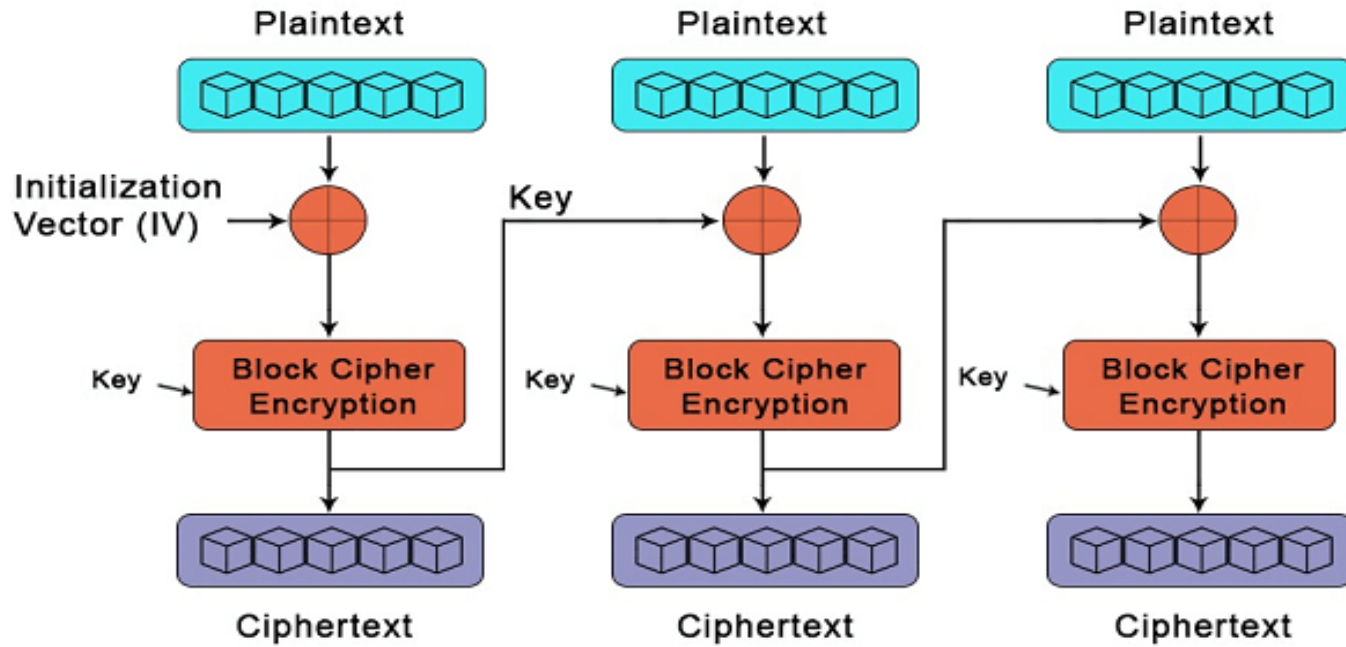
Unit 2: Cryptography

Block Cipher Principles

- Block ciphers are a fundamental type of symmetric encryption algorithm used in modern cryptography. They operate on fixed-size blocks of plaintext and encrypt them into blocks of ciphertext using a secret key.
- The process involves several key principles:

Unit 2: Cryptography

Block Cipher



Unit 2: Cryptography

Block Cipher Principles

- **Block Size:**
- A block cipher operates on fixed-size blocks of plaintext data. The block size is a fixed number of bits, typically 64 or 128 bits, and the plaintext message is divided into these blocks before encryption.
- The block size is an essential parameter in block ciphers, as it determines the amount of data that can be processed in each encryption operation.

Unit 2: Cryptography

Block Cipher Principles

- **Key Size:**
- The key size refers to the length of the secret key used in the block cipher algorithm. The larger the key size, the stronger the encryption provided by the block cipher.
- The key size is also a critical factor in determining the security of the cipher. A longer key size increases the complexity of brute-force attacks, where an attacker tries all possible keys to decrypt the ciphertext.

Unit 2: Cryptography

Block Cipher Principles

- **Rounds:**
- Block ciphers typically use multiple rounds of transformations to process the plaintext blocks and generate ciphertext blocks.
- Each round consists of a series of mathematical operations, such as substitution, permutation, and mixing, applied to the input data using the secret key. These operations create confusion and diffusion, making it harder for attackers to decipher the encryption without the key.

Unit 2: Cryptography

Block Cipher Principles

- **Substitution and Permutation:**
- Substitution and permutation are two fundamental operations used in block ciphers to achieve confusion and diffusion.
- Substitution involves replacing elements in the plaintext block with elements from a substitution table (also known as an S-box). This process introduces non-linearity into the encryption, making it resistant to algebraic attacks.

Unit 2: Cryptography

Block Cipher Principles

- **Substitution and Permutation:**
- Permutation, also called transposition, involves rearranging the elements in the block according to a specific pattern. This operation ensures that each bit of the plaintext influences multiple bits in the ciphertext, increasing the diffusion of information.

Unit 2: Cryptography

Block Cipher Principles

- **Key Expansion:**
- The secret key used in a block cipher is often shorter than the block size. To use the key in each round, it needs to be expanded into a set of round keys, one for each round of the encryption process.
- Key expansion algorithms generate round keys from the original secret key, ensuring that each round uses a unique key that is derived from the main secret key.

Unit 2: Cryptography

Block Cipher Principles

- **Confusion and Diffusion:**
- These are two essential principles in modern block cipher design.
- Confusion aims to make the relationship between the plaintext and the ciphertext as complex as possible, so even a small change in the plaintext will produce a significantly different ciphertext.

Unit 2: Cryptography

Block Cipher Principles

- **Confusion and Diffusion:**
- Diffusion ensures that each bit of the plaintext affects multiple bits of the ciphertext, spreading the influence of the plaintext throughout the entire ciphertext.

Unit 2: Cryptography

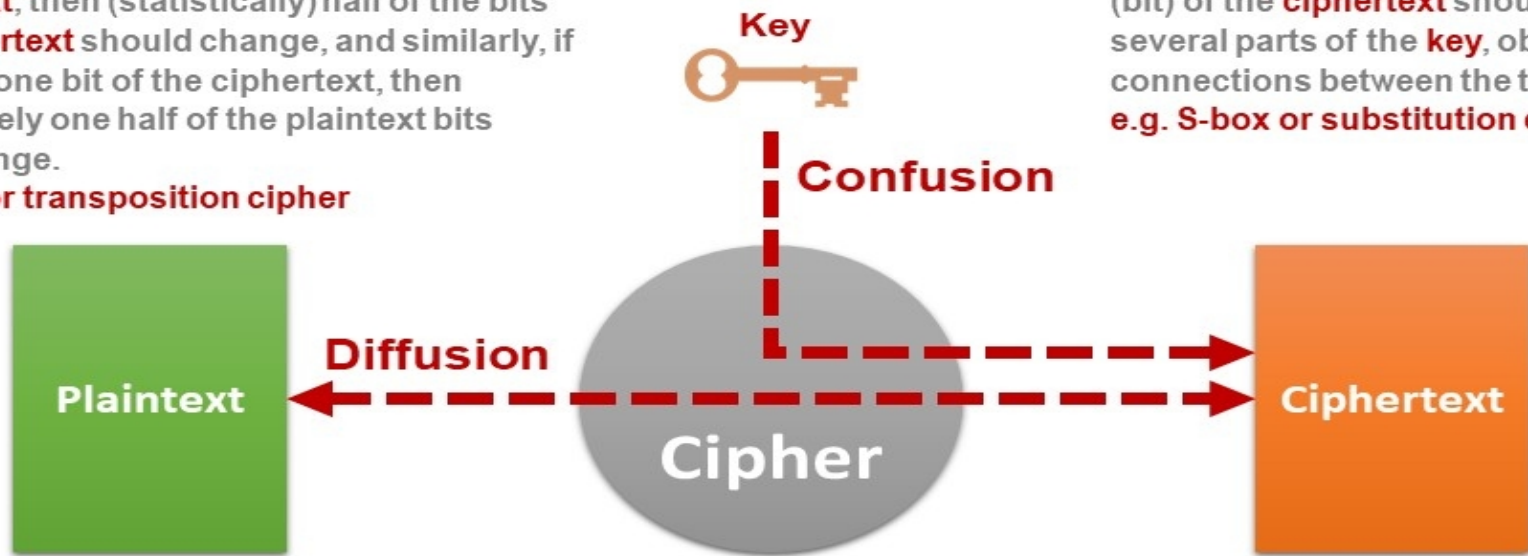
Confusion and Diffusion

Diffusion means that if we change a single bit of the **plaintext**, then (statistically) half of the bits in the **ciphertext** should change, and similarly, if we change one bit of the ciphertext, then approximately one half of the plaintext bits should change.

e.g. P-box or transposition cipher

Confusion means that each binary digit (bit) of the **ciphertext** should depend on several parts of the **key**, obscuring the connections between the two.

e.g. S-box or substitution cipher



Unit 2: Cryptography

Block Cipher Principles

- By combining these principles, block ciphers provide secure encryption of fixed-size blocks of data and are widely used in various encryption protocols, secure communication channels, and data protection mechanisms.
- The security of block ciphers depends on the complexity of their design, the strength of the key, and the resistance to known attacks. Well-known block ciphers include AES (Advanced Encryption Standard), DES (Data Encryption Standard), and 3DES (Triple DES), BlowFish, TwoFish, Serpent.

Unit 2: Cryptography

DES (Data Encryption Standard)

- DES, which stands for Data Encryption Standard, is a symmetric encryption algorithm that was widely used for secure data transmission and storage. It was developed in the 1970s by IBM and later standardized by the National Institute of Standards and Technology (NIST) in 1977. DES was highly influential in the field of cryptography but is now considered outdated due to its small key size, which makes it vulnerable to brute-force attacks.

Unit 2: Cryptography

DES (Data Encryption Standard)

- **Key Generation:**
- DES uses a 64-bit key.
- However, 8 of those bits are used for parity, leaving only 56 bits for actual encryption, meaning there are 2^{56} possible keys.
- The user's input key is processed to generate total 16 subkeys (size of 48-bit) used in the encryption process. Each subkey is derived from the original 56-bit key through a process called key scheduling.

Unit 2: Cryptography

DES (Data Encryption Standard)

- **Key Generation:**
- Keys are used to change the end result of an encryption process in such a way that it wouldn't be possible for an attacker to predict the plaintext just by knowing the algorithm.
- DES begins with one key, which is used to form sub keys that are applied during each 'round'. To produce the final ciphertext, the DES algorithm is run 16 times, and each time it is run is called a round with the output of one round acting as the input for the next.

Unit 2: Cryptography

DES (Data Encryption Standard)

- **Key Generation:**
- Now, the first thing we do is choose a random 64 bit number to be our key K. Mind you, both the sender and the receiver need to have the same 64 bit key to communicate.
- For each round, one subkey is needed. This subkey is unique to that round, and is essentially a permutation of our master key K that we set at the beginning. A total of 16 subkeys are generated, one for each round. They're denoted as K1 to K16.

Unit 2: Cryptography

DES (Data Encryption Standard)

- **Key Generation:**
- The first step to derive our round keys is to permute the key according to the Permuted Choice-1 (PC-1) table. We shall rearrange the elements of our data block using the following PC1 table.

Unit 2: Cryptography

DES (Data Encryption Standard)

- Key Generation:

C						
57	49	41	33	25	17	09
01	58	50	42	34	26	18
10	02	59	51	43	35	27
19	11	03	60	52	44	36
D						
63	55	47	39	31	23	15
07	62	54	46	38	30	22
14	06	61	53	45	37	29
21	13	13	28	20	12	04

PC1

Unit 2: Cryptography

DES (Data Encryption Standard)

- **Key Generation:**
- The “C” and “D” are the "Left" and "Right" halves of the data block. The numbers given in the table decide which bits from the input key belong to which (left or right) sections of the state of the key scheduling.
- During the PC-1 permutation, each bit of the original key is rearranged and placed at a new position in accordance with the table. Since the first cell of table "C" reads 57, the 57th bit from the original key will be placed in the first cell.

Unit 2: Cryptography

DES (Data Encryption Standard)

- **Key Generation:**
- Looking at the table and the key, you may have noticed that they are only 56 bits in length, rather than 64 bits. This is because every eighth bit (8, 16, 24, 32, 40, 48, 56) is specified for use as parity bits to ensure that the key has been received correctly. Due to these parity bits, the DES algorithm has the security of a 56 bit key in practice.
- Our 56 bit key is now split into left and right halves, each 28 bits in length. We will now be using this for our next step.

Unit 2: Cryptography

DES (Data Encryption Standard)

- **Key Generation:**
- The next step is shifting the key to the left either by 1 or 2 bits depending on the round number.
- For example, if we're in the 5th encryption round and had 1101101101101101 as our key, then according to the table we'll have to left shift it by 2 spaces. Hence after this round, our key will now be: 0110110110110111.

No. of round	No. of left rotations
1	1
2	1
3	2
4	2
5	2
6	2
7	2
8	2
9	1
10	2
11	2
12	2
13	2
14	2
15	2
16	1

Unit 2: Cryptography

DES (Data Encryption Standard)

- **Key Generation:**
- Hence, sixteen different subkeys are generated, with one for each round of the DES process.
- Next, we permute the key according to the Permuted Choice-2 table as given. Here we once again rearrange the bits in our key according to the table below.

Unit 2: Cryptography

DES (Data Encryption Standard)

- Key Generation:

14	17	11	24	01	05
03	28	15	06	21	10
23	19	12	04	26	08
16	07	27	20	13	02
41	52	31	37	47	55
30	40	51	45	33	48
44	49	39	56	34	53
46	42	50	36	29	32

PC2

Unit 2: Cryptography

DES (Data Encryption Standard)

- **Key Generation:**
- Similar to the previous permutation, the key is shuffled in such a way that the new position of the key bit is derived as indicated by the table.
- Hence the 14th bit of the key is placed at the first position of the new permuted key, the 17th bit of the key is placed at the second position of the new permuted key..so on and so forth.

Unit 2: Cryptography

DES (Data Encryption Standard)

- **Key Generation:**
- On further inspection, we can see that the newly generated key has only 48 bits, unlike the previous keys that generated 56 bits. This process is called compression permutation.
- The top half of the table contains numbers from 1 to 28, and the bottom half contains numbers from 29 to 56. This ensures the left and right halves of the resulting subkeys are separate.

Unit 2: Cryptography

DES (Data Encryption Standard)

- **Encryption Process:**
- DES operates on 64-bit blocks of plaintext at a time.
- If the input message is not a multiple of 64 bits, padding is applied to make it a complete block.
- The initial permutation (IP) is applied to the 64-bit plaintext block.
- The plaintext block is divided into two 32-bit halves, L0 and R0.

Unit 2: Cryptography

58	50	42	34	26	18	10	02
60	52	44	36	28	20	12	04
62	54	46	38	30	22	14	06
64	56	48	40	32	24	16	08
57	49	41	33	25	17	09	01
59	51	43	35	27	19	11	03
61	53	45	37	29	21	13	05
63	55	47	39	31	23	15	07

Initial Permutation Table

Unit 2: Cryptography

DES (Data Encryption Standard)

- **Encryption Process:**
- Initial Permutation Function:
- Just like what we did with the key, the plaintext is broken into blocks of 64 bits each. The plaintext is the data that needs to be encrypted and transmitted from sender to receiver. It is also padded if the last block is not the required 64 bits long.
- Then, it is fed to the initial permutation function, which confuses the text before any operation is done on it. The IP (Initial Permutation) does not contain any cryptographic value.

Unit 2: Cryptography

DES (Data Encryption Standard)

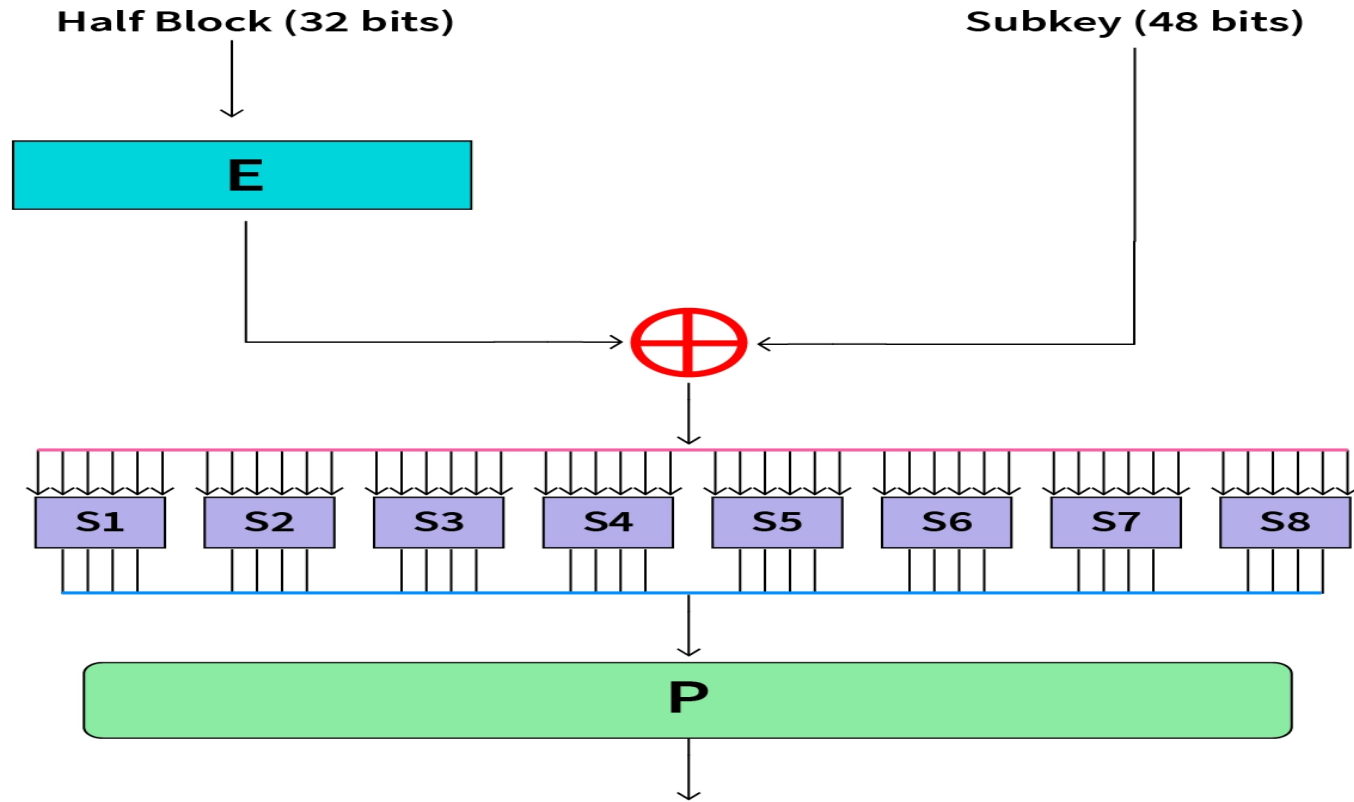
- **Encryption Process:**
- The output of this step is taken as input for the next step:
- Splitting the Blocks:
- Once the IP process is finished, the data is split into 2 halves, a left block L0 made-up of 32 bits, and a right block R0, made up of 32 bits.

Unit 2: Cryptography

DES (Data Encryption Standard)

- **Encryption Process:**
- After the splitting of the blocks, we move to the next function. In round one, the right half of the block is taken and undergoes the following steps:
 - A)Expansion Permutation (E)
 - B)Key mixing ($\oplus$)
 - C)Substitution (S1, S2,...,S8)
 - D)Permutation (P)

Unit 2: Cryptography

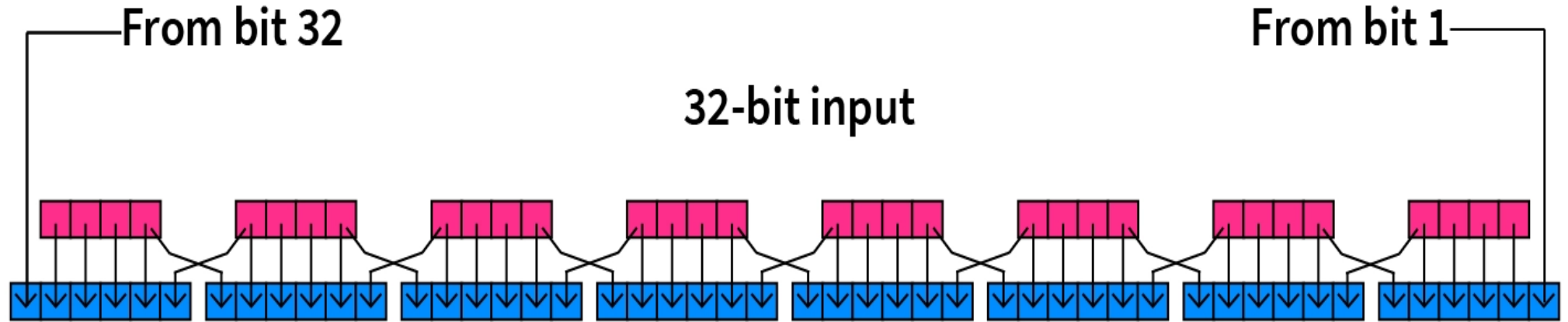


Unit 2: Cryptography

DES (Data Encryption Standard)

- **A) Expansion Permutation (E):**
- Three things are accomplished by the expansion permutation, an important one being the avalanche effect, where one bit of input data directly affects the output of 2 different bits. This step also makes sure the right half has 48 bits, so that it has the same length as the subkey for the upcoming step.
- Another important outcome of the Expansion Permutation is that it permits the data to be squeezed together in the substitution operation by making the output longer than the input.

Unit 2: Cryptography



Unit 2: Cryptography

DES (Data Encryption Standard)

- **A) Expansion Permutation (E):**
- Next, the blocks are permuted once again using the following expansion function table:

32	01	02	03	04	05
04	05	06	07	08	09
08	09	10	11	12	13
12	13	14	15	16	17
16	17	18	19	20	21
20	21	22	23	24	25
24	25	26	27	28	29
28	29	30	31	32	01

E

Unit 2: Cryptography

DES (Data Encryption Standard)

- **B) Key Mixing ($\oplus$):**
- Once we have expanded the block to 48 bits, we apply the subkey of the 1st round which we derived from the key scheduling we did before. The block is then amended by the subkey using the XOR table.
- For example, if the last 8 bits of the expanded key is 01001011, then the key after the key mixing step becomes: 10110100.

Unit 2: Cryptography

DES (Data Encryption Standard)

- **B) Key Mixing ($\oplus$):**

Input		Output
A	B	A xor B
0	0	0
0	1	1
1	0	1
1	1	0

Unit 2: Cryptography

DES (Data Encryption Standard)

- **C) Substitution (S):**
- Substitution is used to make the data more complex so that it cannot be deciphered easily. 8 pre-made tables called substitution boxes or S-Boxes are used to transform each 6 bit input into a 4 bit output.

Unit 2: Cryptography

DES (Data Encryption Standard)

- C) Substitution (S):

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
--	---	---	---	---	---	---	---	---	---	---	----	----	----	----	----	----

S₁

0	14	4	13	1	2	15	11	8	3	10	6	12	5	9	0	7
1	0	15	7	3	14	2	13	1	10	6	12	11	9	5	3	8
2	4	1	14	8	13	6	2	11	15	12	9	7	13	10	5	0
3	15	12	8	2	4	9	1	7	5	11	3	14	10	0	6	13

S₂

0	15	4	8	14	6	11	3	4	9	7	2	13	12	0	5	10
1	3	13	4	7	15	2	8	14	12	0	1	10	6	9	11	5
2	0	14	7	11	10	4	13	1	5	8	12	6	9	3	2	15
3	13	8	10	1	3	15	4	2	11	6	7	12	0	5	14	9

S₃

0	10	0	9	14	6	3	15	5	1	13	12	7	11	4	2	8
1	13	7	0	9	3	4	6	10	2	8	5	14	12	11	15	1
2	13	6	4	9	8	15	3	0	11	1	2	12	5	10	14	7
3	1	10	13	0	6	9	8	7	4	15	14	3	11	5	2	12

S₄

0	7	13	14	3	0	6	9	10	1	2	8	5	11	12	4	15
1	13	8	11	5	6	15	0	3	4	7	2	12	1	10	14	9
2	10	6	9	0	12	11	7	13	15	1	3	14	5	2	8	4
3	3	15	0	6	10	1	13	8	9	4	5	11	12	7	2	14

S₅

0	2	12	4	1	7	10	11	6	8	5	3	15	13	0	14	9
1	14	11	2	12	4	7	13	1	5	0	15	10	3	9	8	6
2	4	2	1	11	10	13	7	8	15	9	12	5	6	3	0	14
3	11	8	12	7	1	14	2	13	6	15	0	9	10	4	5	3

S₆

0	12	1	10	15	9	2	6	8	0	13	3	4	14	7	5	11
1	10	15	4	2	7	12	9	5	6	1	13	14	0	11	3	8
2	9	14	15	5	2	8	12	3	7	0	4	10	1	13	11	6
3	4	3	2	12	9	5	15	10	11	14	1	7	6	0	8	13

S₇

0	4	11	2	14	15	0	8	13	3	12	9	7	5	10	6	1
1	13	0	11	7	4	9	1	10	14	3	5	12	2	15	8	6
2	1	4	11	13	12	3	7	14	10	15	6	8	0	5	9	2
3	6	11	13	8	1	4	10	7	9	5	0	15	14	2	3	12

S₈

0	13	2	8	4	6	15	11	1	10	9	3	14	5	0	12	7
1	1	15	13	8	10	3	7	4	12	5	6	11	0	14	9	2
2	7	11	4	1	9	12	14	2	0	6	10	13	15	3	5	8
3	2	1	14	7	4	10	8	13	15	12	9	0	3	5	6	11

Unit 2: Cryptography

DES (Data Encryption Standard)

- **Feistel Network:**
- DES uses a Feistel network structure. In each round (16 rounds in total), a function is applied to the right half (R) while the left half (L) remains unchanged.
- The output of the function is then XORed with the left half (L) to produce the new right half (R).
- The left and right halves are then swapped for the next round.

Unit 2: Cryptography

DES (Data Encryption Standard)

- **Feistel Function:**
- The Feistel function takes a 32-bit right half (R) and a 48-bit subkey as input.
- The 32-bit R is expanded to 48 bits using an expansion permutation.
- The expanded R is XORed with the 48-bit subkey.

Unit 2: Cryptography

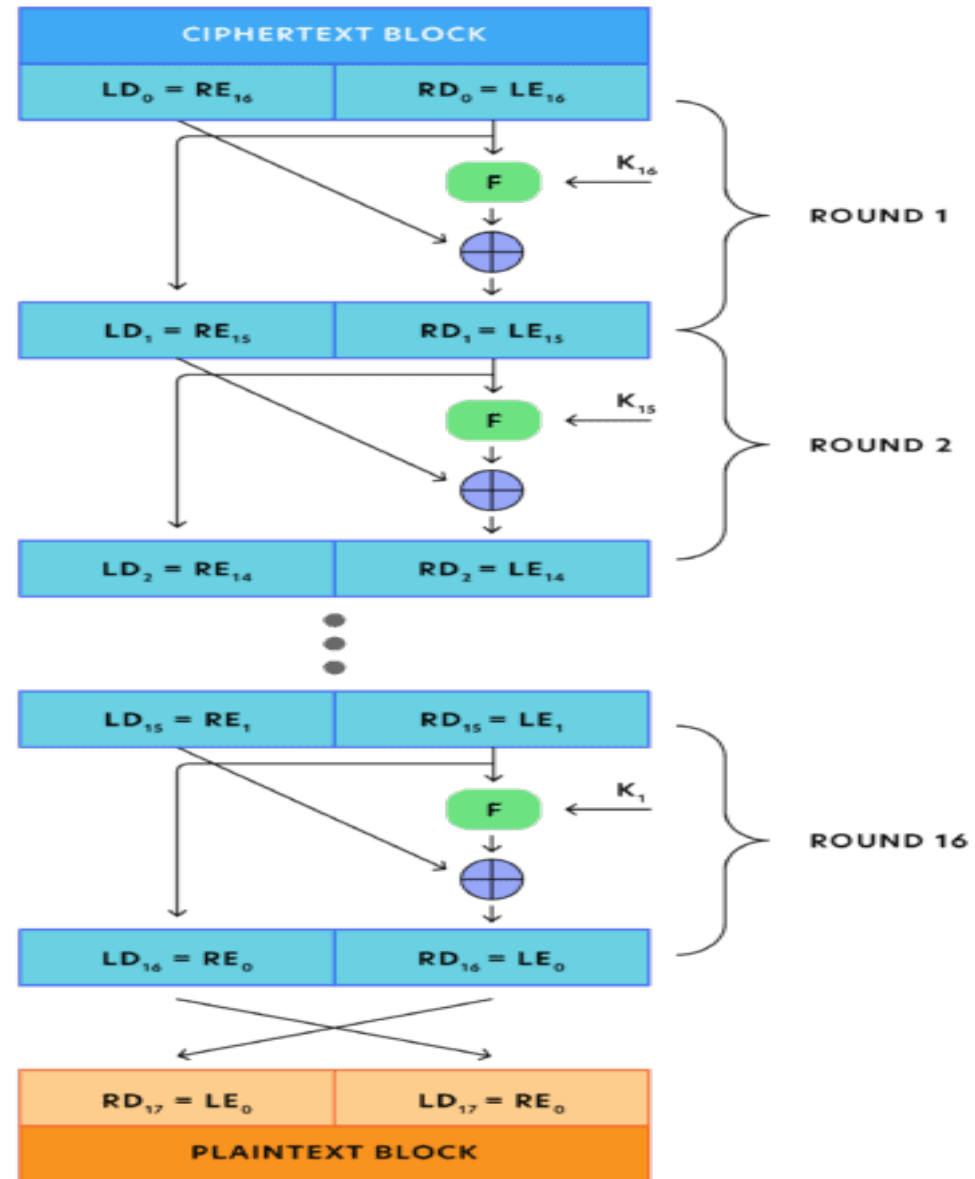
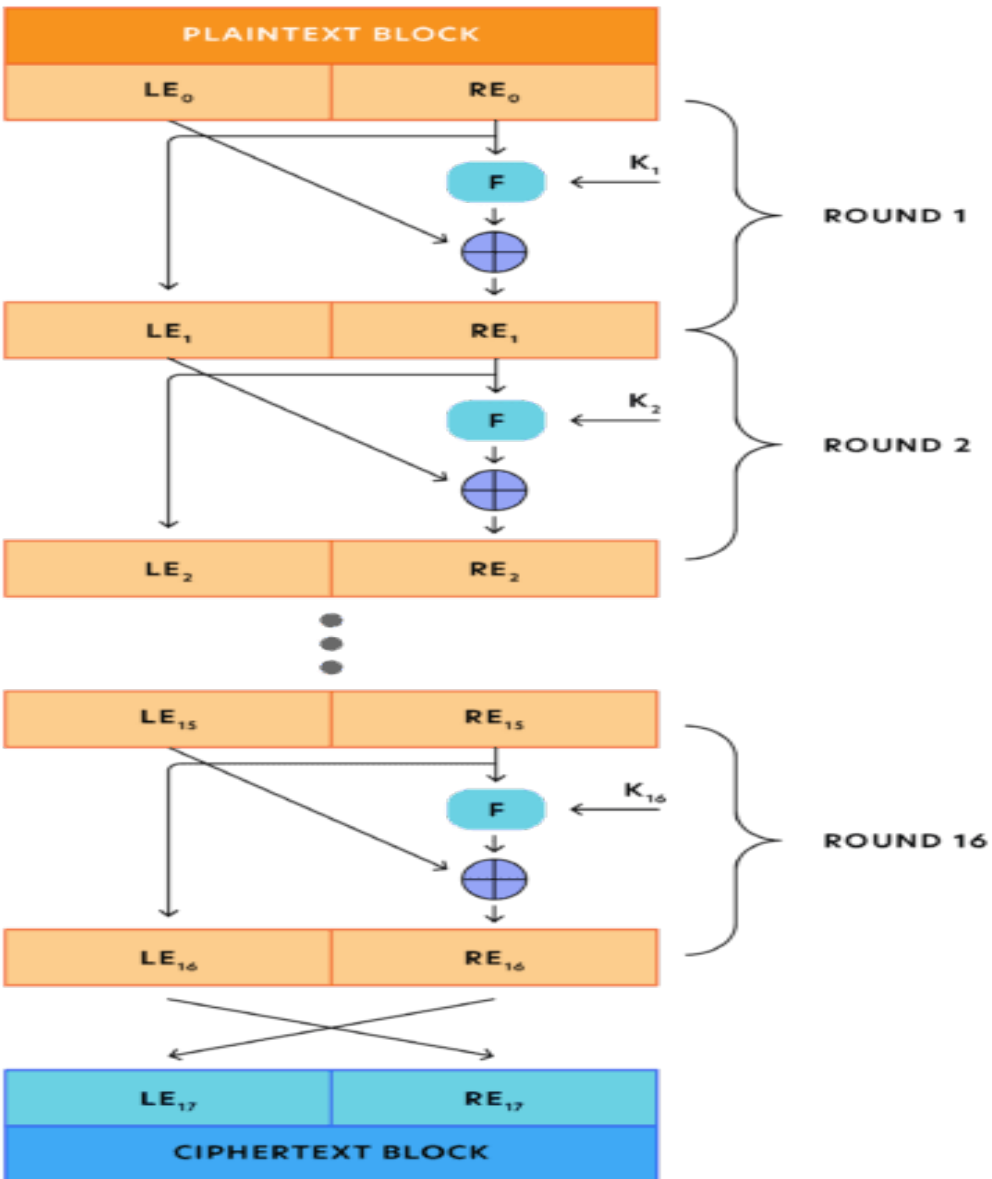
DES (Data Encryption Standard)

- **Feistel Function:**
- The result is then divided into eight 6-bit blocks, which are passed through eight S-boxes (Substitution boxes).
- Each S-box replaces the 6-bit input with a 4-bit output based on a fixed substitution table.
- The outputs from the S-boxes are combined to form a 32-bit block, which is then permuted using a fixed permutation (P-box).

Unit 2: Cryptography

DES (Data Encryption Standard)

- **Decryption:**
- Decryption in DES involves using the same algorithm as encryption but with the subkeys applied in reverse order (i.e., starting with the last subkey and going backwards).
- After the 16 rounds, the final ciphertext goes through an inverse initial permutation (IP^{-1}), and the result is the plaintext.



Unit 2: Cryptography

AES (**A**dvanced **E**ncryption **S**tandard)

- AES stands for Advanced Encryption Standard, and it is a widely used symmetric encryption algorithm. Symmetric encryption means that the same key is used for both encryption and decryption.
- AES is known for its security and efficiency, and it's used to protect sensitive data in various applications, including secure communications, data storage, and more.

Unit 2: Cryptography

AES (Advanced Encryption Standard)

- **Block Cipher:**

AES is a block cipher, which means it operates on fixed-size blocks of data. AES works on blocks of 128 bits (16 bytes). This means that if you have a message longer than 128 bits, you need to break it into blocks and encrypt each block separately.

Unit 2: Cryptography

AES (Advanced Encryption Standard)

- **Key Lengths:**

AES supports key lengths of 128, 192, or 256 bits. The key length determines the strength of the encryption, with longer keys providing higher security. AES-128 is generally considered secure for most practical purposes, while AES-256 offers the highest level of security.

Unit 2: Cryptography

AES (Advanced Encryption Standard)

- **Rounds:**

AES encryption consists of multiple rounds of transformation.

- **The number of rounds depends on the key length:** 10 rounds for AES-128, 12 rounds for AES-192, and 14 rounds for AES-256. In each round, a series of operations are performed on the data.

Unit 2: Cryptography

AES (Advanced Encryption Standard)

- **SubBytes:**

In this step, each byte in the block is replaced with a corresponding byte from an S-box (Substitution Box). This non-linear substitution adds confusion to the data.

- **ShiftRows:**

In this step, the bytes in each row of the block are shifted by different amounts. This diffusion ensures that data doesn't stay in one place.

Unit 2: Cryptography

AES (Advanced Encryption Standard)

- **MixColumns:**

In this step, the columns of the block are mixed by applying a mathematical transformation to each column. This further spreads the data.

- **AddRoundKey:**

In this step, a round-specific subkey is XORed (bitwise exclusive OR) with the block. The subkey is derived from the main encryption key.

Unit 2: Cryptography

AES (Advanced Encryption Standard)

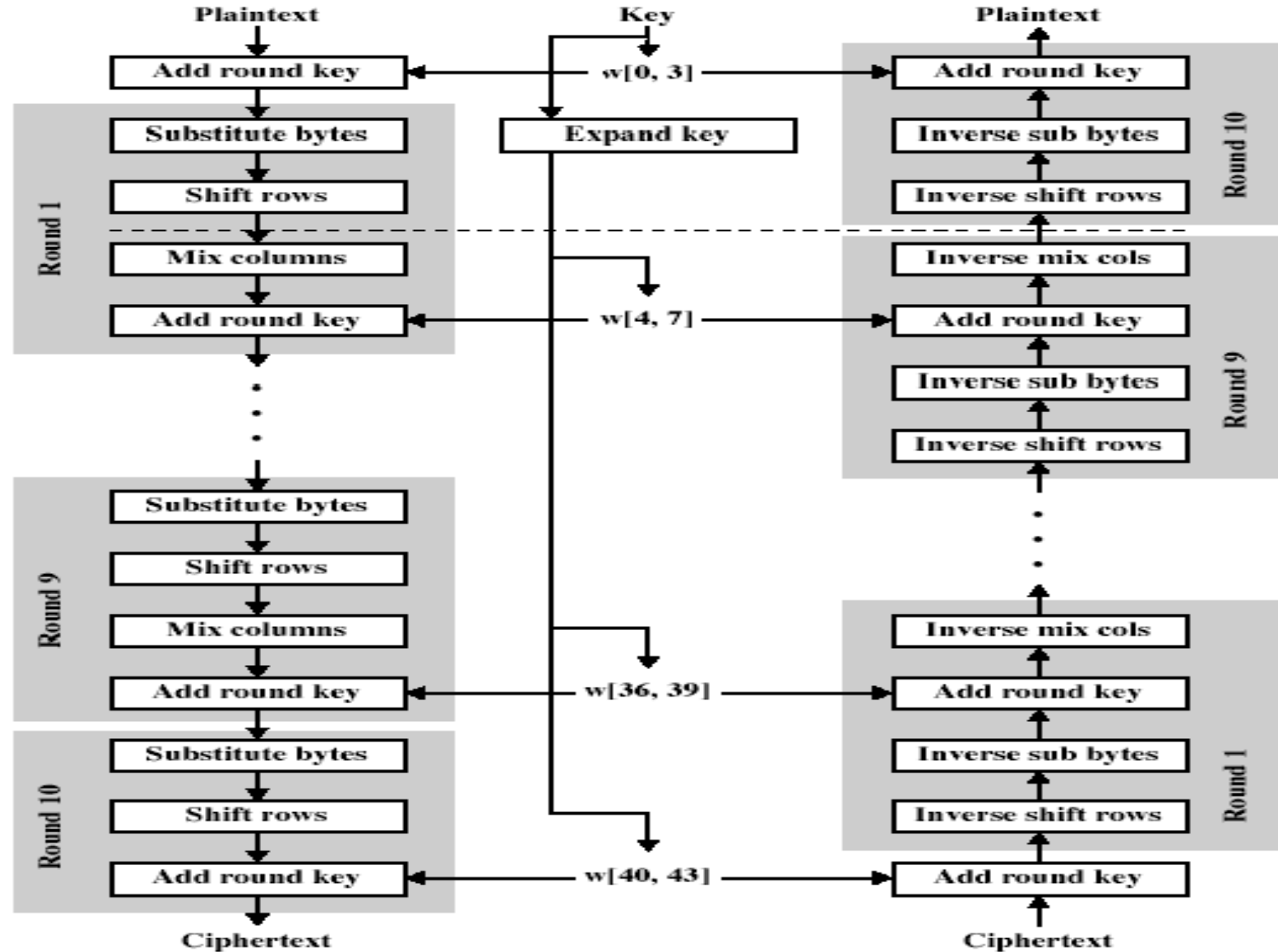
- **Key Expansion:** To generate the round keys for each round, AES uses a key expansion algorithm. This algorithm derives a set of round keys from the original encryption key.
- **Decryption:** AES decryption is the reverse process of encryption. The same round keys are used in reverse order, starting with the last round key. The decryption process consists of the following steps in reverse order: AddRoundKey, MixColumns (only in the last round of decryption), ShiftRows, and SubBytes.

Unit 2: Cryptography

AES (Advanced Encryption Standard)

- **Security:** AES is considered highly secure when used with an appropriate key length. Its security is based on the difficulty of various mathematical operations, including substitution, permutation, and XOR operations, making it resistant to various cryptographic attacks.
- **Modes of Operation:** AES can be used in different modes of operation to encrypt data of arbitrary lengths, such as Electronic Codebook (ECB), Cipher Block Chaining (CBC), and others. Each mode has its own way of handling multiple blocks of data.

AES - 128



Unit 2: Cryptography

Attacks on DES & AES

- Both DES (Data Encryption Standard) and AES (Advanced Encryption Standard) are widely used symmetric encryption algorithms, but they differ significantly in terms of security. Here's an overview of attacks on both algorithms:

Unit 2: Cryptography

Attacks on DES

- **Brute Force Attack:** DES is vulnerable to brute force attacks due to its small key size (56 bits). In a brute force attack, an attacker tries every possible key until the correct one is found. Given today's computing power, a dedicated attacker can break DES in a reasonable amount of time.
- **Known-Plaintext Attack:** If an attacker has access to a sufficient amount of known plaintext and corresponding ciphertext pairs encrypted with the same key, they can use this information to deduce the key.

Unit 2: Cryptography

Attacks on DES

- **Differential Cryptanalysis:** This is a statistical attack that analyzes the differences between pairs of plaintexts and their corresponding ciphertexts. DES is somewhat susceptible to this type of attack.
- Differential cryptanalysis is a general form of cryptanalysis applicable primarily to block ciphers, but also to stream ciphers and cryptographic hash functions.
- In the broadest sense, it is the study of how differences in information input can affect the resultant difference at the output.

Unit 2: Cryptography

Attacks on DES

- **Linear Cryptanalysis:** Linear cryptanalysis is another statistical attack that aims to find linear relationships between the plaintext, ciphertext, and key bits. While DES was designed to resist such attacks, there are still known linear relationships that can be exploited.
- In cryptography, linear cryptanalysis is a general form of cryptanalysis based on finding affine approximations to the action of a cipher. Attacks have been developed for block ciphers and stream ciphers.

Unit 2: Cryptography

Attacks on DES

- There are two parts to **linear cryptanalysis**.
- The first is to construct linear equations relating plaintext, ciphertext and key bits that have a high bias; that is, whose probabilities of holding (over the space of all possible values of their variables) are as close as possible to 0 or 1.
- The second is to use these linear equations in conjunction with known plaintext - ciphertext pairs to derive key bits.

Unit 2: Cryptography

Attacks on DES

- Linear cryptanalysis is newer than differential cryptanalysis. DES is more vulnerable to linear cryptanalysis than to differential cryptanalysis.
- S-boxes are not very resistant to linear cryptanalysis. It has been shown that DES can be broken using 2^{43} pairs of known plaintexts. However, from the practical point of view, finding so many pairs is very unlikely.

Unit 2: Cryptography

Attacks on DES

- **Men-in-the-Middle Attack:** This attack involves encrypting a known plaintext with all possible keys and storing the results. Then, decrypting a known ciphertext with all possible keys and searching for matches in the stored results. DES's 56-bit key length makes this attack feasible.

Due to these vulnerabilities and advances in computing power, DES is no longer considered secure for modern cryptographic purposes. AES was developed to address these security concerns.

Unit 2: Cryptography

Attacks on AES

- **Brute Force Attack:** AES is considered secure against brute force attacks due to its larger key sizes. AES-128, AES-192, and AES-256 have key lengths of 128, 192, and 256 bits, respectively. The larger key sizes make exhaustive search attacks computationally infeasible with current technology.
- **Known-Plaintext and Chosen-Plaintext Attacks:** AES has withstood extensive cryptanalysis, and no practical attacks against it have been discovered in this regard. Even with access to known or chosen plaintexts, AES remains highly secure.

Unit 2: Cryptography

Attacks on AES

- **Differential and Linear Cryptanalysis:** AES was designed to resist both differential and linear cryptanalysis, and extensive research has shown that it provides strong security against these attacks.

Unit 2: Cryptography

Attacks on AES

- **Side-Channel Attacks:** While AES itself is strong, implementations can be vulnerable to side-channel attacks, where an attacker observes physical characteristics of the encryption process (e.g., power consumption or electromagnetic radiation) to deduce the key. Protecting against these attacks requires secure implementation practices.

Unit 2: Cryptography

Attacks on DES & AES

- In summary, DES is vulnerable to various attacks due to its small key size, while AES offers significantly improved security with larger key sizes and resistance to known attacks. AES is widely considered secure for most practical purposes and continues to be a fundamental Symmetric encryption algorithm in use today.

Unit 2: Cryptography

Modes of Operation

- Block ciphers encrypt fixed size blocks
eg. DES encrypts 64-bit blocks, with 56-bit key
- Need way to use in practice, given usually have arbitrary amount of information to encrypt

Unit 2: Cryptography

Modes of Operation

- There are five modes intended to cover virtually all the possible applications of encryptions for which DES is used. These modes determine how blocks of plaintext are transformed into blocks of ciphertext.
- (1) Electronic Codebook (ECB)
- (2) Cipher Block Chaining (CBC)
- (3) Cipher Feedback (CFB)
- (4) Output Feedback (OFB)
- (5) Counter (CTR)

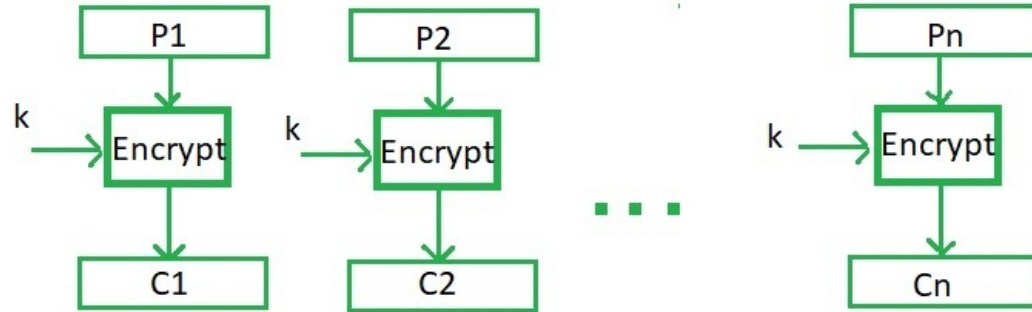
Unit 2: Cryptography

Modes of Operation

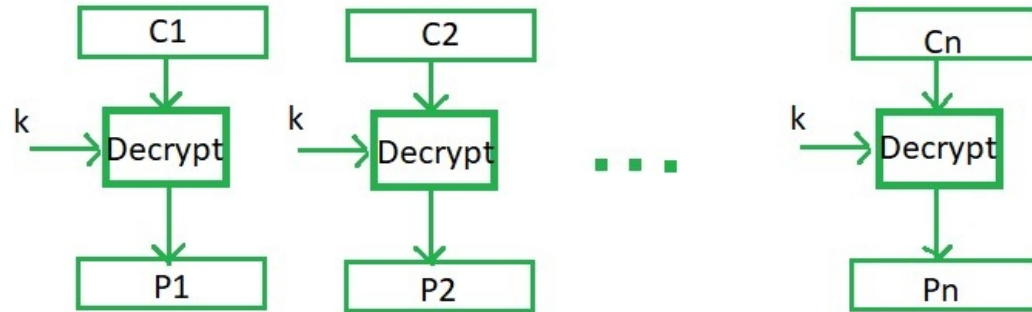
- **Electronic Codebook (ECB):**
- In ECB mode, each block of plaintext is encrypted independently using the same encryption key.
- The same plaintext block will always produce the same ciphertext block.
- It does not provide confidentiality for patterns in the data, as identical plaintext blocks yield identical ciphertext blocks.

Unit 2: Cryptography

Encryption



Decryption



Unit 2: Cryptography

Modes of Operation

- **Electronic Codebook (ECB):**

Advantages of using ECB:

- Parallel encryption of blocks of bits is possible, thus it is a faster way of encryption.
- Simple way of the block cipher.

Disadvantages of using ECB:

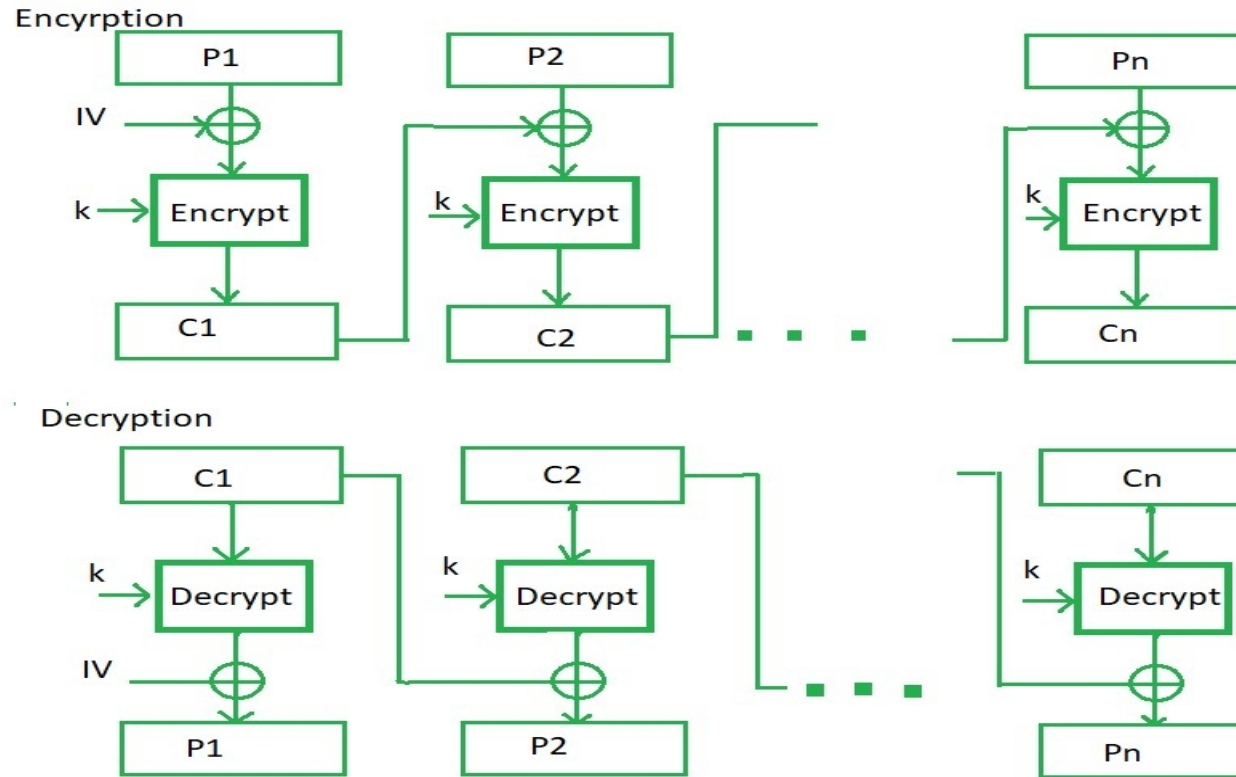
- Prone to cryptanalysis since there is a direct relationship between plaintext and ciphertext.

Unit 2: Cryptography

Modes of Operation

- **Cipher Block Chaining (CBC):**
- In CBC mode, each plaintext block is XORed with the ciphertext of the previous block before encryption.
- This chaining of blocks adds an element of feedback and ensures that identical plaintext blocks will produce different ciphertext blocks.
- CBC is more secure than ECB and provides confidentiality and integrity.

Unit 2: Cryptography



Unit 2: Cryptography

Modes of Operation

- **Cipher Block Chaining (CBC):**

Advantages of CBC:

- CBC works well for input greater than b bits.
- CBC is a good authentication mechanism.
- Better resistive nature towards cryptanalysis than ECB.

Disadvantages of CBC:

- Parallel encryption is not possible since every encryption requires a previous cipher.

Unit 2: Cryptography

Modes of Operation

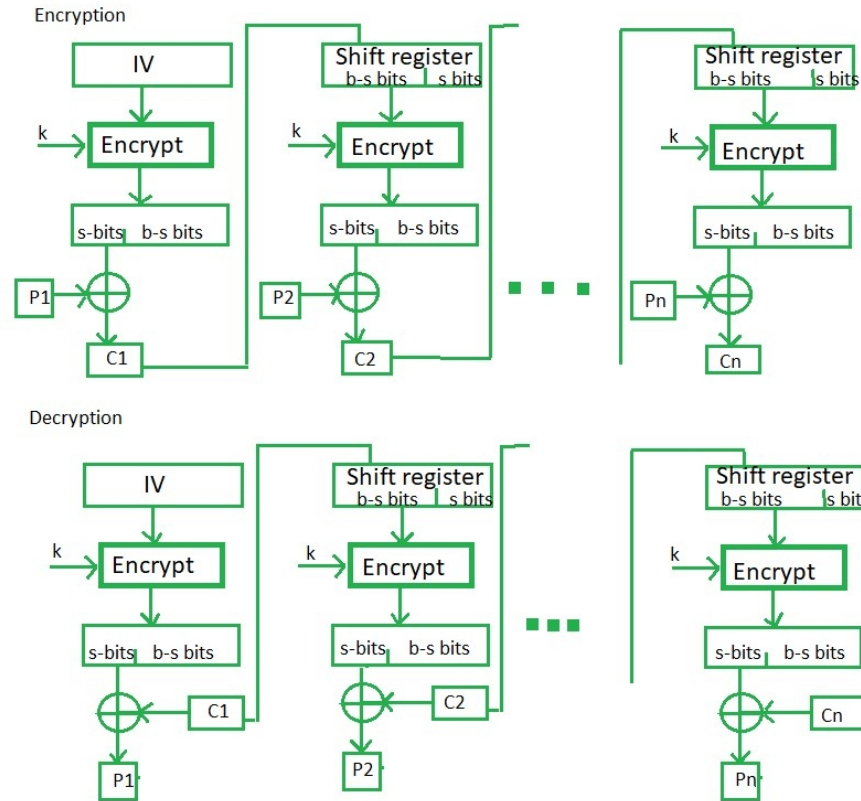
- **Cipher Feedback (CFB):**
- CFB mode turns a block cipher into a stream cipher, where the encryption of each block produces a pseudo-random keystream.
- The keystream is XORed with the plaintext to produce ciphertext.
- It allows for the encryption of individual bits or bytes, not just blocks.

Unit 2: Cryptography

Modes of Operation

- **Cipher Feedback (CFB):**
- It provides error propagation, meaning if a bit is altered in the ciphertext, it will affect the corresponding bit in the plaintext.

Unit 2: Cryptography



Unit 2: Cryptography

Modes of Operation

- **Cipher Feedback (CFB):**

Advantages of CFB:

- Since, there is some data loss due to the use of shift register, thus it is difficult for applying cryptanalysis.

Disadvantages of using CFB:

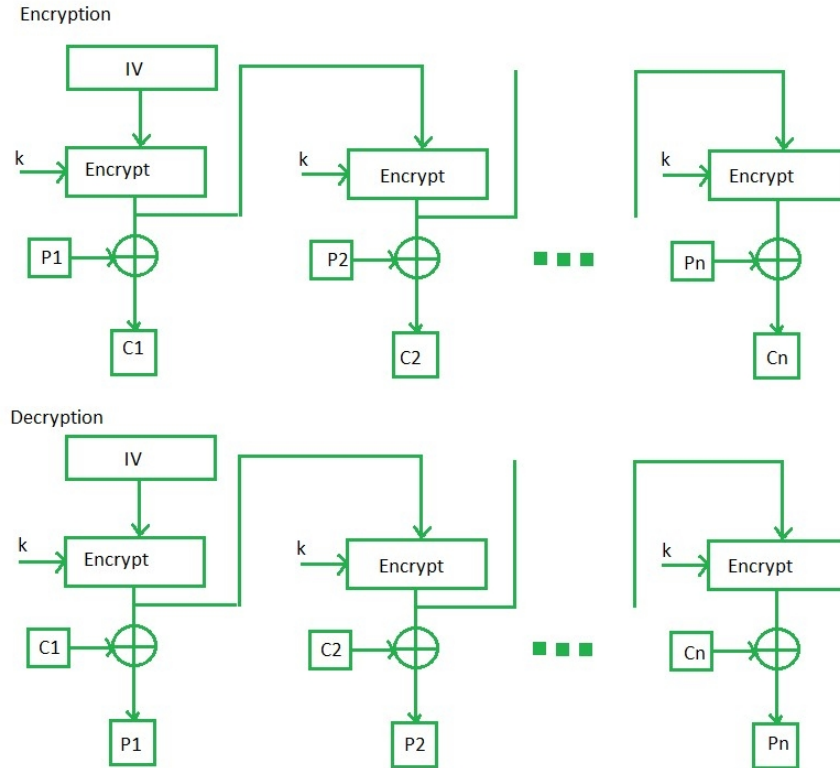
- The drawbacks of CFB are the same as those of CBC mode. Both block losses and concurrent encryption of several blocks are not supported by the encryption. Decryption, however, is parallelizable and loss-tolerant.

Unit 2: Cryptography

Modes of Operation

- **Output Feedback (OFB):**
- OFB mode also converts a block cipher into a stream cipher.
- It generates a keystream, which is independent of the plaintext, and XORs it with the plaintext to produce ciphertext.
- Like CFB, it allows for the encryption of individual bits or bytes.
- Errors in the ciphertext do not affect the decryption process, which can be an advantage in some situations.

Unit 2: Cryptography



Unit 2: Cryptography

Modes of Operation

- **Output Feedback (OFB):**

Advantages of OFB:

- In the case of CFB, a single bit error in a block is propagated to all subsequent blocks. This problem is solved by OFB as it is free from bit errors in the plaintext block.

Disadvantages of OFB:

- The drawback of OFB is that, because to its operational modes, it is more susceptible to a message stream modification attack than CFB.

Unit 2: Cryptography

Modes of Operation

- **Counter (CTR):**
- CTR mode turns a block cipher into a stream cipher by using a counter as the input for each block.
- The counter value is encrypted, and the resulting keystream is XORed with the plaintext.
- CTR mode is highly parallelizable, making it efficient for hardware and software implementations.
- It does not require error propagation, and errors in the ciphertext do not affect the decryption process.

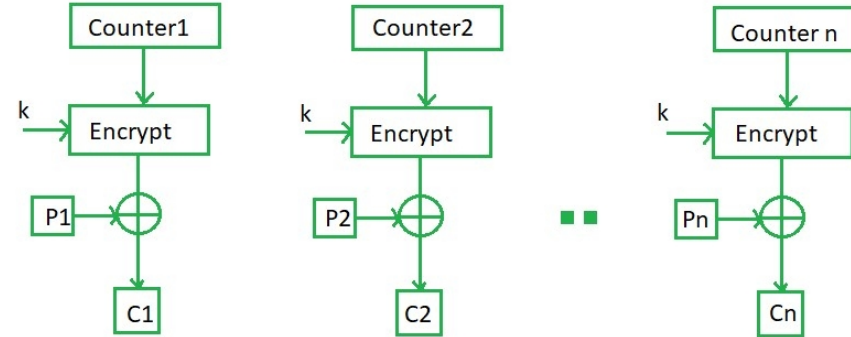
Unit 2: Cryptography

Modes of Operation

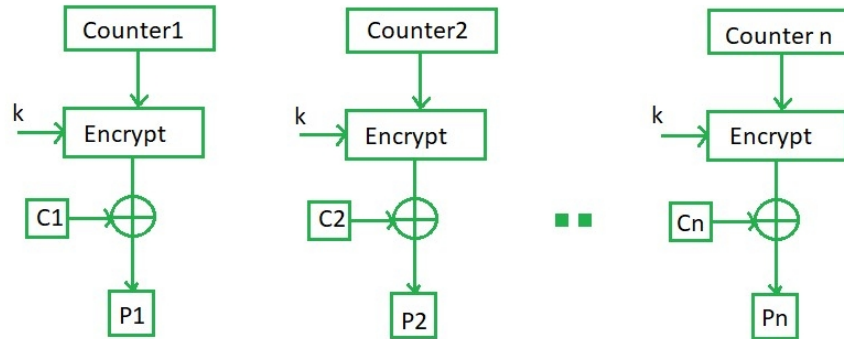
- **Counter (CTR):**
- Each mode has its own strengths and weaknesses, and the choice of mode depends on the specific requirements of the encryption application, such as security, parallelization, and error handling. CBC, CFB, and OFB are considered to provide better security than ECB, especially when dealing with multiple blocks of data, while CTR mode is often chosen for its efficiency in certain scenarios.

Unit 2: Cryptography

Encryption



Decryption



Unit 2: Cryptography

Modes of Operation

- **Counter (CTR):**

Advantages of Counter:

- Since there is a different counter value for each block, the direct plaintext and ciphertext relationship is avoided. This means that the same plain text can map to different ciphertext.
- Parallel execution of encryption is possible as outputs from previous stages are not chained as in the case of CBC.

Unit 2: Cryptography

Modes of Operation

- **Counter (CTR):**

Disadvantages of Counter:

- The fact that CTR mode requires a synchronous counter at both the transmitter and the receiver is a severe drawback. The recovery of plaintext is erroneous when synchronisation is lost.

Unit 2: Cryptography

Stream Cipher

- In stream cipher, one byte is encrypted at a time while in block cipher ~128 bits are encrypted at a time.
- Initially, a key(k) will be supplied as input to pseudorandom bit generator and then it produces a random 8-bit output which is treated as keystream.
- The resulted keystream will be of size 1 byte, i.e., 8 bits.

Unit 2: Cryptography

Stream Cipher

- Stream Cipher follows the sequence of pseudorandom number stream.
- One of the benefits of following stream cipher is to make cryptanalysis more difficult, so the number of bits chosen in the Keystream must be long in order to make cryptanalysis more difficult.
- By making the key more longer it is also safe against brute force attacks.

Unit 2: Cryptography

Stream Cipher

- The longer the key the stronger security is achieved, preventing any attack.
- Keystream can be designed more efficiently by including more number of 1s and 0s, for making cryptanalysis more difficult.
- Considerable benefit of a stream cipher is, it requires few lines of code compared to block cipher.

Unit 2: Cryptography

Stream Cipher

- **Encryption:**
- For Encryption, Plain Text and Keystream produces Cipher Text (Same keystream will be used for decryption).
- The Plaintext will undergo XOR operation with keystream bit-by-bit and produces the Cipher Text.
- Plain Text: 1 0 0 1 1 0 0 1
- Keystream: 1 1 0 0 0 0 1 1
- Cipher Text : 0 1 0 1 1 0 1 0

Unit 2: Cryptography

Stream Cipher

- **Decryption:**
- For Decryption, Cipher Text and Keystream gives the original Plain Text (Same keystream will be used for encryption).
- The Ciphertext will undergo XOR operation with keystream bit-by-bit and produces the actual Plain Text.
- Cipher Text: 0 1 0 1 1 0 1 0
- Keystream: 1 1 0 0 0 0 1 1
- Plain Text: 1 0 0 1 1 0 0 1

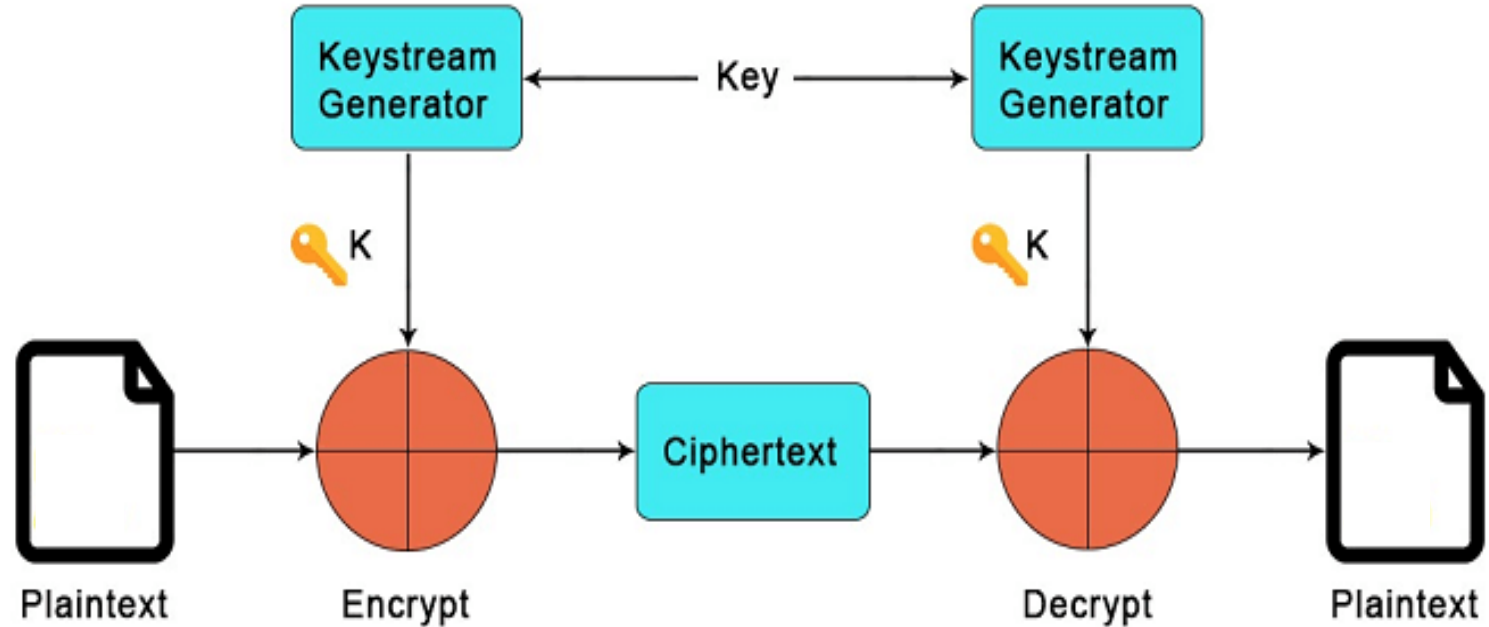
Unit 2: Cryptography

Stream Cipher

Stream Cipher

Decryption is just the reverse process of Encryption.

i.e. performing XOR with Cipher Text.



Unit 2: Cryptography

RC4 Algorithm

- RC4 means Rivest Cipher 4 invented by Ron Rivest in 1987 for RSA Security. It is a Stream Ciphers. Stream Ciphers operate on a stream of data byte by byte.
- RC4 stream cipher is one of the most widely used stream ciphers because of its simplicity and speed of operation.
- It is a variable key-size stream cipher with byte-oriented operations. It uses either 64 bit or 128-bit key sizes. It is generally used in applications such as Secure Socket Layer (SSL), Transport Layer Security (TLS), and also used in IEEE 802.11 wireless LAN std.

Unit 2: Cryptography

RC4 Algorithm

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Unit 2: Cryptography

Types of RC4 Algorithm

- **SPRITZ:** Spritz can be used to build a cryptographic hash function, a deterministic random bit generator (DRBG), an encryption algorithm that supports authenticated encryption with associated data (AEAD).
- **RC4A:** Souraduyti Paul and Bart Preneel have proposed an RC4 variant, which they call RC4A, which is stronger than RC4.
- **VMPC:** VMPC is another variant of RC4 which stands for Variably Modified Permutation Composition.

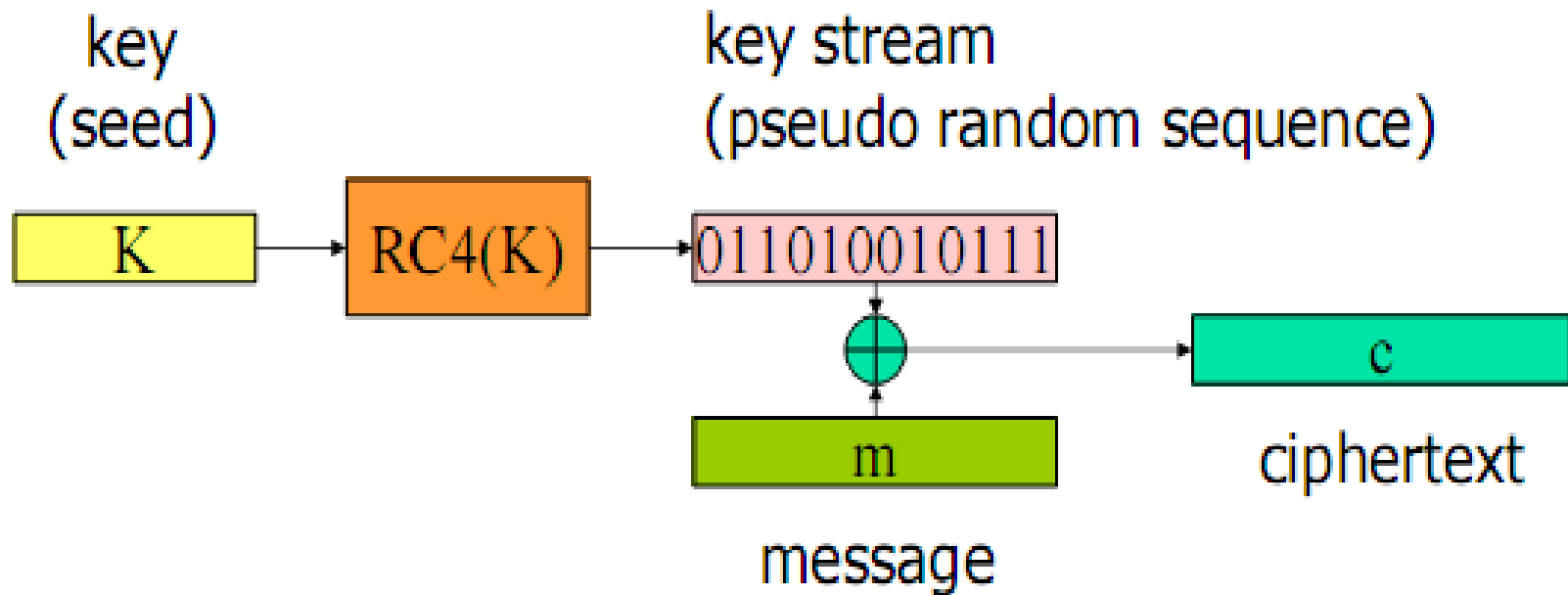
Unit 2: Cryptography

Types of RC4 Algorithm

- **RC4A+:** RC4A+ is a modified version of RC4 with a more complex three-phase key schedule which takes about three times as long as RC4 and a more complex output function which performs four additional lookups in the S array for each byte output, taking approximately 1.7 times as long as basic RC4.

Unit 2: Cryptography

RC4 Algorithm



Unit 2: Cryptography

Working of RC4 Algorithm

- **Encryption Procedure**

- The user inputs a plain text file and a secret key.
- The encryption engine then generates the keystream by using KSA and PRGA Algorithm.
- This keystream is now XOR with the plain text, this XORing is done byte by byte to produce the encrypted text.
- The encrypted text is then sent to the intended receiver, the intended receiver will then decrypted the text and after decryption, the receiver will get the original plain text.

Unit 2: Cryptography

Working of RC4 Algorithm

- **Decryption Procedure**
- Decryption is achieved by doing the same byte-wise X-OR operation on the Ciphertext.
- **Example:** Let A be the plain text and B be the keystream $(A \text{ xor } B) \text{ xor } B = A$

Unit 2: Cryptography

Advantages of RC4 Algorithm

- RC4 stream ciphers are simple to use.
- The speed of operation in RC4 is fast as compared to other ciphers.
- RC4 stream ciphers are strong in coding and easy to implement.
- RC4 stream ciphers do not require more memory.
- RC4 stream ciphers are implemented on large streams of data.

Unit 2: Cryptography

Disadvantages of RC4 Algorithm

- If RC4 is not used with strong MAC(Message Authentication Code) then encryption is vulnerable to a bit-flipping attack. (In a bit-flipping attack, an attacker intercepts and modifies the ciphertext in such a way that specific bits are flipped or changed. If the attacker knows the structure of the plaintext or has some knowledge about the content, they can make targeted changes in the ciphertext to produce a predictable change in the decrypted plaintext. This can be especially effective if the encryption method is not secure against such attacks.)

Unit 2: Cryptography

Disadvantages of RC4 Algorithm

- RC4 stream ciphers do not provide authentication.
- RC4 algorithm requires additional analysis before including new systems.
- RC4 stream ciphers cannot be implemented on small streams of data.

Unit 2: Cryptography

Disadvantages of RC4 Algorithm

- RC4 fails to discard the beginning of output keystream or fails to use non-random or related keys for the algorithm. (One of the significant weaknesses of RC4 is that its initial output keystream bytes can exhibit biases and patterns, making them less random than they should be. To mitigate this issue, it's recommended to discard a certain number of initial keystream bytes before using the rest for encryption. This process is known as "key setup" and involves generating a longer initial keystream and discarding it to ensure a more random starting point.)

Unit 2: Cryptography

Asymmetric Key Cryptography

- Asymmetric key cryptography, also known as public-key cryptography, is a cryptographic technique that uses a pair of keys for secure communication and data encryption.
- This pair of keys consists of a public key and a private key, which are mathematically related but perform distinct roles in the encryption and decryption process.
- Asymmetric cryptography is widely used for various purposes, including secure data transmission, digital signatures, and authentication.

Unit 2: Cryptography

Working of Asymmetric Key Cryptography

- **Key Pair Generation:**
 - **Public Key:** This key is intended to be widely distributed and is used for encryption by anyone who wants to send an encrypted message to the owner of the public key. It is generated by the recipient and can be freely shared.
 - **Private Key:** This key is kept secret by the owner and is used for decrypting messages or verifying digital signatures. Only the owner should have access to the private key.

Unit 2: Cryptography

Working of Asymmetric Key Cryptography

- **Encryption:**
- To send a secure message to the owner of a public key, the sender obtains the recipient's public key.
- The sender uses the recipient's public key to encrypt the message.
- The encrypted message can only be decrypted by the recipient's private key.

Unit 2: Cryptography

Working of Asymmetric Key Cryptography

- **Decryption:**
- The recipient uses their private key to decrypt the encrypted message sent by the sender.
- Since only the recipient has access to the private key, only they can read the original message.

Unit 2: Cryptography

Working of Asymmetric Key Cryptography

- **Digital Signatures:**
- Asymmetric cryptography is also used for digital signatures, which provide data integrity and authentication.
- To sign a message, the sender uses their private key to create a digital signature.
- Anyone with access to the sender's public key can verify the authenticity of the digital signature, ensuring that the message was indeed signed by the private key holder and that it has not been tampered with.

Unit 2: Cryptography

Advantages of Asymmetric Key Cryptography

- **Security:** Asymmetric encryption is considered secure because the private key is not shared or transmitted, reducing the risk of key compromise.
- **Authentication:** Digital signatures help verify the identity of the sender.
- **Key Exchange:** Asymmetric cryptography solves the problem of securely exchanging keys for symmetric encryption.

Unit 2: Cryptography

Advantages of Asymmetric Key Cryptography

- However, asymmetric cryptography is computationally more intensive than symmetric cryptography, making it slower for encrypting and decrypting data. Therefore, it's common to use a combination of both symmetric and asymmetric cryptography in secure communication systems, where asymmetric encryption is primarily used for key exchange and initial authentication, while symmetric encryption is used for the actual data transmission to achieve better performance.

Unit 2: Cryptography

Principles of Public Key Cryptosystem

- **Key Pair:** In a public key cryptosystem, a pair of keys is used: a public key and a private key. These keys are mathematically related but serve different purposes. The public key is freely distributed and used for encryption, while the private key is kept secret and used for decryption or digital signatures.
- **Asymmetry:** Public key cryptosystems rely on mathematical algorithms that make it computationally infeasible to derive the private key from the public key. This asymmetry ensures that even if the public key is widely known, the private key remains secure.

Unit 2: Cryptography

Principles of Public Key Cryptosystem

- **Encryption:** The public key is used to encrypt data or messages. Anyone with access to the recipient's public key can encrypt a message for them. However, only the recipient, who possesses the corresponding private key, can decrypt the message.
- **Decryption:** The private key is used for decrypting data that has been encrypted with the corresponding public key. Only the owner of the private key can perform this operation, ensuring the confidentiality of the message.

Unit 2: Cryptography

Principles of Public Key Cryptosystem

- **Digital Signatures:** Public key cryptosystems enable the creation of digital signatures. A sender can use their private key to sign a message, which proves the authenticity and integrity of the message. Anyone with access to the sender's public key can verify the signature.
- **Key Distribution:** Public keys can be widely distributed and shared openly. This simplifies the process of key distribution compared to symmetric key systems, where a secret key must be securely shared between parties.

Unit 2: Cryptography

Principles of Public Key Cryptosystem

- **Secure Key Generation:** The generation of key pairs must be done securely to prevent key compromise. This typically involves using strong random number generators and secure key storage.
- **Trust Model:** Trust is established in the authenticity of public keys through digital certificates and a public key infrastructure (PKI). Certificate authorities (CAs) validate the association between a public key and an entity's identity, enhancing trust in the public key.

Unit 2: Cryptography

Principles of Public Key Cryptosystem

- **Secure Algorithms:** Public key cryptosystems use mathematically sound and secure algorithms, such as RSA, ECC (Elliptic Curve Cryptography), and DSA (Digital Signature Algorithm). The security of the system relies on the difficulty of mathematical problems, such as factoring large numbers or solving discrete logarithms.
- **Key Management:** Proper key management practices are essential to maintaining the security of public key cryptosystems. This includes key rotation, revocation mechanisms, and secure storage of private keys.

Unit 2: Cryptography

Principles of Public Key Cryptosystem

- **Interoperability:** Public key cryptosystems are designed to work across different platforms and systems, allowing for secure communication between parties with different hardware and software configurations.
- **Performance Considerations:** Public key cryptography can be computationally intensive compared to symmetric key cryptography. Therefore, it's often used in combination with symmetric cryptography for optimal performance and security.

Unit 2: Cryptography

Principles of Public Key Cryptosystem

- These principles collectively enable secure communication, data protection, authentication, and digital signatures in a public key cryptosystem. Public key cryptography plays a crucial role in securing data and online transactions in the modern digital age.

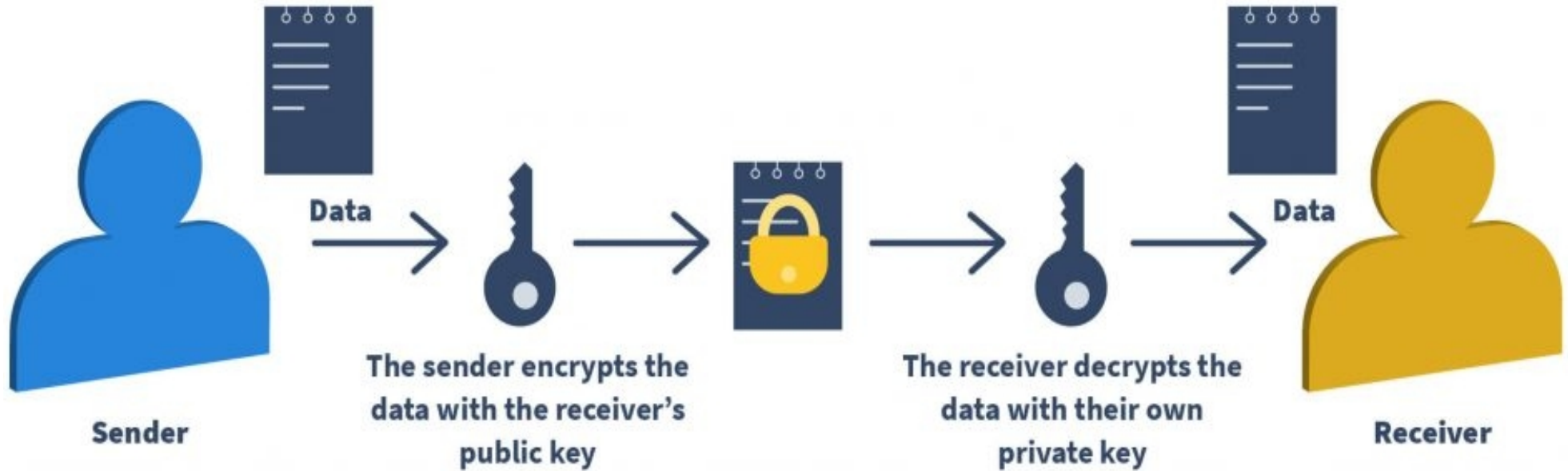
Unit 2: Cryptography

RSA Algorithm

- The RSA (Rivest–Shamir–Adleman) algorithm is a widely used public-key cryptosystem that is used for secure data transmission and digital signatures. It was invented by Ron Rivest, Adi Shamir, and Leonard Adleman in 1977. RSA is based on the mathematical properties of prime numbers and relies on the difficulty of factoring the product of two large prime numbers.

Unit 2: Cryptography

RSA Algorithm



Unit 2: Cryptography

RSA Algorithm

- **Generating the keys:**
- **Select Two Prime Numbers (p and q):**
- Choose two distinct prime numbers, p and q. These should be kept secret. (Let's choose $p = 11$ and $q = 13$)
- **Calculate the Modulus (n):**
- Calculate the modulus (n) by multiplying the two prime numbers:
- $n = p * q$
- ($n = 11 * 13 = 143$)

Unit 2: Cryptography

RSA Algorithm

- **Generating the keys:**
- **Calculate the Totient Function ($\phi(n)$):**
- Calculate the totient function ($\phi(n)$) for the modulus n :
- $\phi(n) = (p - 1) * (q - 1)$
- $(\phi(n) = (11-1) * (13-1) = 10 * 12 = 120)$

Unit 2: Cryptography

RSA Algorithm

- **Generating the keys:**
- **Select an Encryption Exponent (e):**
- Choose an encryption exponent (e) that is coprime (relatively prime) to $\phi(n)$. Typically, e is a small prime number (often 65537 or 3), but it should meet the following conditions:
- $1 < e < \phi(n)$
- $\gcd(e, \phi(n)) = 1$, where gcd denotes the greatest common divisor.
- (e = 7 for this example)

Unit 2: Cryptography

RSA Algorithm

- **Generating the keys:**
- **Calculate the Private Key Exponent (d):**
- Calculate the private key exponent (d) as the modular multiplicative inverse of e modulo $\phi(n)$. In other words, find d such that:
- $e * d \equiv 1 \pmod{\phi(n)}$
- You can use the Extended Euclidean Algorithm to find d.
- (In this case, d=103 because: $7 * 103 \equiv 1 \pmod{120}$)

Unit 2: Cryptography

RSA Algorithm

- **Generating the keys:**
- **Public Key and Private Key Formation:**
- The pair (n, e) forms the public key, which you can share with anyone who wants to encrypt messages for you.
- The pair (n, d) forms the private key, which you must keep secret and use for decrypting messages.
- Public Key: $(n, e) = (143, 7)$
- Private Key: $(n, d) = (143, 103)$

Unit 2: Cryptography

RSA Algorithm

- **Encryption:**
- **Message Preparation:** The sender prepares the message they want to send. This message is typically represented as a numerical value or a series of numerical values (often bytes or characters converted to numbers).
- **Public Key Encryption:** The sender encrypts the message using the recipient's public key, which consists of two values: (n, e) . This step involves raising the message to the power of e and then taking the modulus n : $\text{Encrypted Message} = \text{Message}^e \bmod n$

Unit 2: Cryptography

RSA Algorithm

- **Encryption:**
- **Sending the Encrypted Message:** The sender sends the encrypted message to the recipient.
- Let's say $m=65$.
- $c \equiv m^e \pmod n$
- $c \equiv 65^7 \pmod{143}$
- $c \equiv 125$

Unit 2: Cryptography

RSA Algorithm

- **Decryption:**
- **Receiving the Encrypted Message:** The recipient receives the encrypted message sent by the sender.
- **Private Key Decryption:** The recipient uses their private key, consisting of two values: (n, d) , to decrypt the message. Decryption involves raising the encrypted message to the power of d and then taking the modulus n : $\text{Decrypted Message} = \text{Encrypted Message}^d \bmod n$
- **Message Recovery:** The recipient obtains the original message from the decrypted result.

Unit 2: Cryptography

RSA Algorithm

- **Decryption:**
- Decrypt c using the private key:
- $m \equiv c^d \pmod{n}$
- $m \equiv 125^{103} \pmod{143}$
- $M \equiv 65$

Unit 2: Cryptography

Attacks on RSA

- **Brute Force Attack:** This is the most straightforward but impractical attack. It involves trying every possible private key until the correct one is found. RSA's security relies on the difficulty of factoring large composite numbers, and factoring large numbers is computationally expensive.
- **Factorization Attack:** This is the primary attack against RSA. If an attacker can factor the modulus (n) into its two prime factors (p and q), they can easily compute the private key. Factoring large semiprime numbers (the product of two large primes) is a challenging problem, but advances in factoring algorithms and increasing computational power have made it a concern.

Unit 2: Cryptography

Attacks on RSA

- **Timing Attack and Side-Channel Attacks:** These are physical attacks that exploit information leaked during the encryption or decryption process, such as the time it takes to perform certain operations or power consumption patterns. These attacks are difficult to execute and require physical access to the hardware.
- **Chosen Plaintext Attack:** In this attack, the attacker can choose specific plaintexts to be encrypted and obtain the corresponding ciphertexts. Over time, by analyzing the patterns and responses, they may gain information about the private key.

Unit 2: Cryptography

Attacks on RSA

- **Adaptive Chosen Ciphertext Attack (CCA2):** This attack involves the attacker submitting ciphertexts to the decryption oracle and obtaining the corresponding plaintexts. The attacker can then use this information to decrypt other ciphertexts.
- **Padding Oracle Attack:** This attack exploits vulnerabilities in the padding scheme used in RSA, such as PKCS#1 v1.5 padding. By sending carefully crafted ciphertexts and observing how the server responds (e.g., with padding error messages), an attacker may gain information about the private key.

Unit 2: Cryptography

Attacks on RSA

- **Common Modulus Attack:** In cases where two different plaintexts are encrypted with the same modulus but different public exponents (e_1 and e_2), and the attacker knows both ciphertexts and e_1 and e_2 , it is possible to compute the greatest common divisor (GCD) of the two ciphertexts. This can reveal information about the plaintexts or the private key.
- **Small Private Exponent Attack:** If the private exponent (d) is chosen poorly or is too small, it may become vulnerable to certain mathematical attacks, such as the Wiener's attack.

Unit 2: Cryptography

Attacks on RSA

- To protect against these attacks and maintain the security of RSA encryption, it's essential to use appropriate key lengths (typically 2048 bits or higher), employ secure padding schemes (e.g., Optimal Asymmetric Encryption Padding (OAEP) for encryption and Probabilistic Signature Scheme (PSS) for signatures), and follow best practices in key management and secure implementation. Additionally, monitoring for any emerging vulnerabilities or cryptographic advancements is crucial to maintaining the security of RSA encryption.

Unit 2: Cryptography

Key Management

- Key management is the process of generating, storing, distributing, and controlling cryptographic keys used in various security systems, such as encryption, authentication, and digital signatures. Effective key management is critical for maintaining the confidentiality, integrity, and authenticity of data and communications in a secure manner.

Unit 2: Cryptography

Key Management

- **Key Generation:** This involves the creation of cryptographic keys using a secure and random process. Keys can be symmetric (same key used for encryption and decryption) or asymmetric (public and private key pairs).
- **Key Storage:** Secure storage of keys is crucial to prevent unauthorized access. Hardware security modules (HSMs), secure key vaults, and secure containers are used to protect keys from theft or tampering.

Unit 2: Cryptography

Key Management

- **Key Distribution:** Keys need to be securely shared with authorized parties. This can involve various methods, such as physically transporting keys or using secure key exchange protocols.
- **Key Rotation:** To enhance security, keys should be rotated regularly. This involves generating new keys and replacing old ones while ensuring uninterrupted service.
- **Key Revocation:** When a key is compromised or no longer needed, it must be revoked to prevent unauthorized access. This requires a process for identifying and deactivating keys.

Unit 2: Cryptography

Key Management

- **Key Escrow:** In some cases, keys may need to be stored in escrow to recover encrypted data in emergencies or legal situations. This should be done with strict controls to prevent misuse.
- **Key Lifecycle Management:** Managing keys throughout their entire lifecycle, from creation to disposal, is essential for maintaining security.
- **Compliance and Auditing:** Key management processes should adhere to relevant security standards and regulations. Regular audits ensure that the key management system is functioning correctly and securely.

Unit 2: Cryptography

Key Management

- **Key Recovery:** In cases where keys are lost or forgotten, a key recovery mechanism may be needed to regain access to encrypted data.
- **Key Destruction:** When keys are no longer needed, they should be securely destroyed to prevent any future use.

Unit 2: Cryptography

Key Management

- Key management is crucial in various security applications, including:
- **Data Encryption:** Protecting sensitive data at rest and in transit by encrypting it with strong cryptographic keys.
- **Authentication:** Verifying the identity of users or systems using digital certificates and keys.
- **Digital Signatures:** Ensuring the authenticity and integrity of digital documents and transactions through the use of private keys.
- **Secure Communication:** Securing communications over the internet, such as in HTTPS connections, using encryption keys.

Unit 2: Cryptography

Key Management

- Key management is crucial in various security applications, including:
- **Tokenization:** Replacing sensitive data with tokens and managing the keys used to map tokens back to the original data.
- **Secure Messaging:** Ensuring the privacy and security of instant messaging and email communications through encryption and digital signatures.

Unit 2: Cryptography

Key Distribution Scenarios

- Key distribution is a critical aspect of cryptographic systems and plays a central role in ensuring secure communication and data protection. There are various scenarios in which key distribution is necessary, each with its own unique requirements and challenges. Here are some key distribution scenarios:
- **Secure Communication Over a Network:**
- **SSL/TLS:** Key distribution in Secure Socket Layer (SSL) or Transport Layer Security (TLS) protocols is essential for securing web communications. Public key certificates are used to establish secure connections.

Unit 2: Cryptography

Key Distribution Scenarios

- **End-to-End Encryption:**
- Messaging Apps: Secure messaging applications like WhatsApp or Signal rely on key distribution to enable end-to-end encryption. Users exchange encryption keys to ensure only the intended recipients can decrypt messages.
- **Virtual Private Networks (VPNs):**
- VPNs use keys to encrypt and decrypt network traffic between a client and a server or between two network endpoints. Keys must be securely distributed to ensure the confidentiality of the data being transmitted.

Unit 2: Cryptography

Key Distribution Scenarios

- **Secure File Sharing:**
- In scenarios where files need to be securely shared among multiple users, a secure key distribution mechanism ensures that only authorized parties can access the files.
- **Wireless Networks:**
- Wi-Fi networks often use pre-shared keys (PSKs) for encryption. These keys need to be distributed securely to all authorized users of the network.

Unit 2: Cryptography

Key Distribution Scenarios

- **Secure Software Updates:**
- Software vendors need to securely distribute updates and patches to their users. This often involves signing updates with a cryptographic key to ensure their authenticity.
- **Secure Remote Access:**
- In remote access scenarios like SSH (Secure Shell) or remote desktop applications, keys are used to authenticate users and secure the communication. Key distribution is essential for this purpose.

Unit 2: Cryptography

Key Distribution Scenarios

- **IoT and Embedded Devices:**

- In the Internet of Things (IoT), secure key distribution is crucial for ensuring the security of devices and the data they collect and transmit.

- **Multi-Party Transactions:**

- In scenarios involving multiple parties, such as financial transactions or blockchain networks, key distribution ensures that only authorized parties can participate and access the shared data.

Unit 2: Cryptography

Key Distribution Scenarios

- **Cloud Services:**
- Cloud providers often use encryption to protect data stored in the cloud. Secure key distribution ensures that customer data remains confidential and can only be decrypted by authorized users or applications.
- **Machine-to-Machine (M2M) Communication:**
- In industrial and IoT applications, machines and devices communicate with each other. Secure key distribution ensures the confidentiality and integrity of the data exchanged.

Unit 2: Cryptography

Diffie Hellman Key Exchange Algorithm

- The Diffie-Hellman key exchange algorithm is a cryptographic protocol that allows two parties to agree on a shared secret key over an insecure communication channel without needing to transmit the key directly. This shared secret key can then be used for secure communication, such as encrypting and decrypting messages. The protocol was developed by Whitfield Diffie and Martin Hellman in 1976 and is widely used in secure communications today.

Unit 2: Cryptography

Diffie Hellman Key Exchange Algorithm Working

- **Setup:**
- Before starting the key exchange, both parties, Alice and Bob, agree on two things:
- A large prime number, denoted as " p ."
- A primitive root modulo " p ," denoted as " g ." A primitive root is a number whose powers modulo " p " generate all possible residues from 1 to " $p-1$."
- These parameters (p and g) can be publicly known and are typically part of a standard, like a security protocol or agreed upon beforehand. They don't need to be kept secret.

Unit 2: Cryptography

Diffie Hellman Key Exchange Algorithm Working

- **Private Keys:**
- Alice and Bob each generate their own private key.
- Alice's private key: a random integer "a."
- Bob's private key: a random integer "b."

Unit 2: Cryptography

Diffie Hellman Key Exchange Algorithm Working

- **Key Calculation:**
- Alice calculates her public key, A:
- $A = (g^a) \bmod p$
- Bob calculates his public key, B:
- $B = (g^b) \bmod p$
- The mod is used to find the remainder when the result of raising "g" to the power of "a" or "b" is divided by "p." This operation ensures that both A and B are relatively small numbers compared to "p."

Unit 2: Cryptography

Diffie Hellman Key Exchange Algorithm Working

- **Exchange Public Keys:**
- Alice and Bob exchange their public keys (A and B) over the insecure communication channel. It's important to note that an eavesdropper can see A and B , but they cannot easily calculate the private keys (a and b) from them due to the discrete logarithm problem, which is the foundation of the security of this protocol.

Unit 2: Cryptography

Diffie Hellman Key Exchange Algorithm Working

- **Shared Secret Calculation:**
- Alice receives B from Bob, and Bob receives A from Alice.
- Alice calculates the shared secret key, S:
- $S = (B^a) \bmod p$
- Bob calculates the shared secret key, S:
- $S = (A^b) \bmod p$

Unit 2: Cryptography

Diffie Hellman Key Exchange Algorithm Working

- **Final Shared Secret:**
- Alice and Bob now both have the same shared secret key, S , without ever having transmitted it directly. This shared secret can be used for encryption and decryption of messages between them.

Unit 2: Cryptography

Diffie Hellman Key Exchange Algorithm Example

- **Step 1:** Alice and Bob get public numbers $P = 23$, $G = 9$
- **Step 2:** Alice selected a private key $a = 4$ and Bob selected a private key $b = 3$
- **Step 3:** Alice and Bob compute public values
Alice: $x = (9^4 \bmod 23) = (6561 \bmod 23) = 6$
Bob: $y = (9^3 \bmod 23) = (729 \bmod 23) = 16$
- **Step 4:** Alice and Bob exchange public numbers

Unit 2: Cryptography

Diffie Hellman Key Exchange Algorithm Example

- **Step 5:** Alice receives public key $y = 16$ and Bob receives public key $x = 6$
- **Step 6:** Alice and Bob compute symmetric keys
Alice: $k_a = y^a \bmod p = 65536 \bmod 23 = 9$
Bob: $k_b = x^b \bmod p = 216 \bmod 23 = 9$
- **Step 7:** 9 is the shared secret.



Private = 5



$(6^5) \text{ MOD } 13$
 $(7776) \text{ MOD } 13$
Public = 2



$(9^5) \text{ MOD } 13$
 $(59049) \text{ MOD } 13$
Shared Secret = 3



Agree upon two numbers:

P Prime Number **13**

G Generator of P **6**

Randomly generate a Private Key

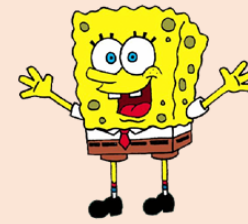
Calculate Public Key:

$(G^{\text{Private}}) \text{ MOD } P$

Exchange Public Keys

Calculate the Shared Secret
 $(\text{Shared Public}^{\text{Private}}) \text{ MOD } P$

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Private = 4



$(6^4) \text{ MOD } 13$
 $(1296) \text{ MOD } 13$
Public = 9



$(2^4) \text{ MOD } 13$
 $(16) \text{ MOD } 13$
Shared Secret = 3

